

APÉRY-LIKE RECURRENCES BEYOND THE RIGID CASE: FOUR-TERM ROWS, $K3$ SURFACES, AND NEW PERIODS

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ABSTRACT. Integral three-term Apéry-like recurrences $(n+1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1}$ have a rank-two Picard–Fuchs operator with four singular points: a rigid situation, classified in the companion paper under the hypothesis that the local system comes from an elliptic surface. We take the first step out of rigidity, to the four-term row $(n+1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1} + R(n)u_{n-2}$, whose operator has rank two with *five* singular points and two accessory parameters. We give the exponent dictionary and the normalisation classes, both in the equal-exponent and in the mixed-exponent case, the Riemann–Hurwitz bound $\deg \mathcal{J} \leq 6c + 4e_3 + 3e_2 - 12 \leq 18$, and the results of complete or box-complete scans. Four things happen at once. *Near-total rigidity persists*: of 2178 integral rows produced, 98.2% have an apparent singularity and are gauge or cusp-move images of three-term rows, and the only one-parameter families found are of that kind. *A finite sporadic list appears*: 30 primitive rows with five genuine singular points, 28 of them with denominator exponent $k = 2$. *$K3$ geometry enters exactly here*: one of the 30 is the Picard–Fuchs system of an elliptic $K3$ with five singular fibres $I_4 I_6 I_3 I_3 IV^*$, which is impossible with four singular points. *New periods appear*: in the mixed-exponent classes there are genuine five-point rows with irrational conjugate singularities whose Apéry limits are $\frac{1}{4}G$, $\frac{1}{2}G - \frac{3}{16}\zeta(2)$, $\frac{3}{8}\zeta(2) - \frac{1}{2}G$ and $L(2, \chi_{-3})$ -combinations, whose gauge partners all give $\Gamma(\frac{1}{4})^4/(64\pi)$, and, one term further, a six-point five-term row whose Apéry limit is the weight-3 cusp-form value $\frac{\sqrt{2}}{3}L(g, 2)$. What does *not* appear is new irrationality-relevant arithmetic: no genuine five-point row has positive score, no unequal-exponent class can contain a decaying row, and the 2-adic content of the new Catalan rows is the ledger’s own.

1. INTRODUCTION

1.1. The rigid case, and one step out. Let

$$(n+1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1}, \quad u_0 = 1, \quad u_{-1} = 0, \quad (1)$$

with P, Q integral of degree ≤ 2 . The generating function $y = \sum u_n t^n$ satisfies a rank-two Fuchsian equation with the four singular points $0, t_1, t_2, \infty$, and such an equation is *rigid*: no accessory parameter, local exponents determining it, and — when the local system comes from an elliptic surface — the surface forced onto Beauville’s and Herfurtners’ finite lists [2, 8]. The companion paper [4] carries this out: its Theorem A is a conditional classification (all four fibres semistable, $\deg \mathcal{J} \leq 12$ by Riemann–Hurwitz and $12 \mid \deg \mathcal{J}$ by a twist-parity lemma, hence $\deg \mathcal{J} = 12$ and a *rational* elliptic surface, with the integral rows exactly Zagier’s seven [16]), its Theorem B collects the unconditional necessary conditions, and the missing converse is Zagier’s conjecture that every integral row has a modular parametrisation — Bombieri–Dwork in rank two with four singular points.

This paper asks what survives one step out. The object is the integral *four-term* row

$$(n+1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1} + R(n)u_{n-2}, \quad u_0 = 1, \quad u_{-1} = u_{-2} = 0, \quad (2)$$

$P = an^2 + bn + c$, $Q = dn^2 + en + f$, $R = gn^2 + hn + j$ integral with $g \neq 0$. Its operator is rank two with *five* singular points $0, t_1, t_2, t_3, \infty$ and therefore has $5 - 3 = 2$ accessory parameters — it is the first non-rigid case. Geometrically the change is just as sharp: elliptic surfaces with four singular fibres form finitely many configurations, while those with five come in one-parameter families (Persson [14]), so the cross-ratio finiteness mechanism of the rigid case disappears. Longer recurrences and their accessory parameters are of course not new — they occur throughout the Apéry-like literature [1, 7, 13] and the accessory-parameter problem for such operators is Dwork’s [3] — but the systematic question of which

integral rows exist, and what they compute, has not been asked one step beyond rank two with four points.

1.2. The answers, in words. Four things happen, and one does not.

- (1) *Near-total rigidity persists.* Of the 2178 distinct integral rows the census produces, 2138 (98.2%) have an *apparent* singularity — a point where the exponent difference is a positive integer and the Frobenius obstruction vanishes, or a half-integral one where the monodromy is $-I$. Their *projective* local systems have only four singular points, so they are gauge or cusp-move images of three-term rows. Every one-parameter family the scan produced is of this kind, and the largest is written down in closed form (Proposition 3.1): the signed binomial transform $v_n = \sum_m \binom{n}{m} (-\nu)^{n-m} u_m$ of a Zagier-class row at a non-characteristic ν .
- (2) *A finite sporadic list appears.* After deduplication, 30 primitive rows with five genuine singular points remain (Table 1); 28 of them have sharp denominator exponent $k = 2$, the four-term echo of the free integration phenomenon, and the other two are Padé-type with $k = 1$. No one-parameter integral family of genuine five-point rows was found.
- (3) *K3 geometry enters exactly here.* With four singular points $\deg \mathcal{J} \leq 12$ always, so no elliptic $K3$ can be the Picard–Fuchs system of a three-term row. With five the bound is 18, and one row of the sporadic list attains $\deg \mathcal{J} = 16$ and is an elliptic $K3$ with fibres $I_4 I_6 I_3 I_3 IV^*$, Picard number 20 and transcendental motive the weight-3 CM newform 32.3.d.a. That row is the subject of the companion note [5]; we state it (§4.4) because it is what the scan found, without repeating its proofs.
- (4) *New periods appear.* The mixed-exponent classes — exponent differences $(0, \frac{1}{2}, \frac{1}{2})$ at the three finite points, which the first census never scanned — contain genuine five-point rows with *irrational conjugate* singular points whose Apéry limits are $\frac{1}{4}G$, $\frac{1}{2}G - \frac{3}{16}\zeta(2)$, $\frac{3}{8}\zeta(2) - \frac{1}{2}G$ and three $L(2, \chi_{-3})$ -combinations, with the whole period matrix classical; the same class in the other half-integral gauge gives $\Gamma(\frac{1}{4})^4/(64\pi)$ on three different rows. One term further still, a six-point five-term row has the weight-3 cusp-form value $\frac{\sqrt{2}}{3}L(g, 2)$ as an honest Apéry limit — the L -value that the $K3$ row itself, having no archimedean limit, can produce only as a fold constant.

What does not happen is arithmetic gain. No genuine five-singular-point row in any scanned class has positive irrationality score $\log(1/|\lambda_2|) - k$; the only positive scores belong to disguised three-term rows with $k = 1$, as in the rigid case. Structurally, a row with $|\lambda_2| < 1$ must have an *irreducible* characteristic cubic (Theorem 5.1), so no unequal-exponent class can contain one, and a *real* such λ_2 forces the cubic to be totally real. The new Catalan-period rows realise the geometric evasion (E1) of the obstruction analysis [6] — irrational conjugate singularities, +0.693 nats of archimedean score over Zagier **E** — but their 2-adic slope set is $\{2\}$ and their limits lie in the ledger’s own $\zeta_2(2)$ and $L_2(2, \chi_{12})$ classes; every one is dominated, as a lattice engine, by the row it would replace.

1.3. Evidence classes. Statements carry one of the markers [PROVED] (an exact argument), [VERIFIED: range] (an exact finite computation over the stated range: exact rational or p -adic arithmetic, or a high-precision computation whose agreement is stated), [NUMERICAL: precision] (a floating-point measurement), [CONJECTURE] and [OPEN], and these are never upgraded. All scans reported are a -block ordered, so every reported box is a complete scan of the stated prefix. Sources, scripts and logs are named in §8.

1.4. Main statements. For orientation we record the three named theorems that organise the paper; each is proved or verified in the section indicated.

Theorem A (the dictionary and its classes). [PROVED] (§2) *The operator of (2) is $t^2\mathcal{R}D^2 + tSD + tV$ with $\mathcal{R}, \mathcal{S}, \mathcal{V}$ as in (3), and its exponents are $(0, 0)$ at $t = 0$, $(0, \rho_i)$ at the roots t_i of $\mathcal{R}$ with $\rho_i = -T(t_i)/(t_i\mathcal{R}'(t_i))$, $T = \mathcal{S} - t\mathcal{R}'$, and $(2 - s_1, 2 - s_2)$ at ∞ with s_1, s_2 the roots of R ; Fuchs reads $\sum \rho_i = s_1 + s_2 - 1$. All three ρ_i coincide if and only if three linear conditions on (b, e, h) hold, and that is automatic when the characteristic cubic is irreducible; the unequal case with a quadratic Galois orbit has the normalisation forms of Theorem 2.9. If the local system is that of an elliptic surface then all exponent differences lie in $\{0, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}\} \bmod 1$ and $\deg \mathcal{J} \leq 6c + 4e_3 + 3e_2 - 12 \leq 18$.*

Theorem B (the census). [VERIFIED: boxes as stated in §3] *Over 27 Kodaira-admissible equal-exponent classes the scan produces 2178 distinct integral rows: 4 fail exact integrality at $n = 120, 2138$ (98.2%) have an apparent singularity and are disguised three-term rows, 6 are rescalings, and 30 are primitive rows with five genuine singular points, exactly one of which passes the $\mathcal{J}$ -map test and is an elliptic K3. The two productive mixed-exponent classes, swept to completion, contain exactly nine primitive genuine five-point rows each, in twist-partner bijection.*

Theorem C (no arithmetic gain). [PROVED] and [VERIFIED: boxes W, PW and the extended window] *A row with $|\lambda_2| < 1$ has an irreducible characteristic cubic, and a real such λ_2 forces it to be totally real; consequently no unequal-exponent class contains a row with $|\lambda_2| < 1$. The complete positive-score window over all 27 equal-exponent classes contains exactly two integral rows, both with $k = 1$ and both carrying an I_0^* at ∞ , hence disguised: no genuine five-singular-point four-term row has a positive score.*

2. THE DICTIONARY AND THE NORMALISATION CLASSES

2.1. The operator. Write $\theta = t d/dt$, $D = d/dt$.

Lemma 2.1 ([PROVED]). $y = \sum_{n \geq 0} u_n t^n$ satisfies $Ly = 0$ with $L = \theta^2 - tP(\theta) + t^2Q(\theta+1) - t^3R(\theta+2)$.

Proof. $[t^{n+1}]\theta^2 y = (n+1)^2 u_{n+1}$, $[t^{n+1}]tP(\theta)y = P(n)u_n$, $[t^{n+1}]t^2Q(\theta+1)y = Q(n)u_{n-1}$ and $[t^{n+1}]t^3R(\theta+2)y = R(n)u_{n-2}$. $\square$

Expanding $\theta = tD$, $\theta^2 = t^2D^2 + tD$ gives $L = t^2\mathcal{R}(t)D^2 + t\mathcal{S}(t)D + t\mathcal{V}(t)$ with

$$\begin{aligned}\mathcal{R}(t) &= 1 - at + dt^2 - gt^3, \\ \mathcal{S}(t) &= 1 - (a+b)t + (3d+e)t^2 - (5g+h)t^3, \\ \mathcal{V}(t) &= -c + (d+e+f)t - (4g+2h+j)t^2.\end{aligned}\tag{3}$$

The singular points are $t = 0$, the three roots t_1, t_2, t_3 of $\mathcal{R}$, and $t = \infty$. Since $t^3(\lambda^3 - a\lambda^2 + d\lambda - g) = \mathcal{R}(1/\lambda)$ with $\lambda = 1/t$, the characteristic roots $\lambda_i = 1/t_i$ are the roots of

$$\chi(\lambda) = \lambda^3 - a\lambda^2 + d\lambda - g,\tag{4}$$

and five distinct singular points requires $g \neq 0$ and $\Delta = 18adg - 4a^3g + a^2d^2 - 4d^3 - 27g^2 \neq 0$.

Theorem 2.2 (exponent dictionary; [PROVED]). *The exponents of L are $(0, 0)$ at $t = 0$ (a MUM point, so a logarithm is forced); $(0, \rho_i)$ at a simple root t_i of $\mathcal{R}$, with*

$$T := \mathcal{S} - t\mathcal{R}' = 1 - bt + (d+e)t^2 - (2g+h)t^3, \quad \rho_i = 1 - \frac{\mathcal{S}(t_i)}{t_i\mathcal{R}'(t_i)} = -\frac{T(t_i)}{t_i\mathcal{R}'(t_i)};$$

and $(2 - s_1, 2 - s_2)$ at $t = \infty$, where s_1, s_2 are the roots of R and $\delta_\infty := s_2 - s_1$. Moreover $\sum_i \rho_i = s_1 + s_2 - 1 = -h/g - 1$.

Proof. The first two are the indicial equations of $t^2\mathcal{R}D^2 + t\mathcal{S}D + t\mathcal{V}$ at $t = 0$ and at a simple zero of $\mathcal{R}$. At ∞ , Lt^μ has leading term $-R(\mu+2)t^{\mu+3}$, so $\mu+2 \in \{s_1, s_2\}$; with the convention (exponent) = $-\mu$ the exponents are $2 - s_i$. For the last claim, $\sum_i \rho_i = -\sum_i \text{Res}_{t_i}(T/(t\mathcal{R}))$; the rational function $T/(t\mathcal{R})$ has residue $T(0)/\mathcal{R}(0) = 1$ at $t = 0$ and behaves like $((2g+h)/g)t^{-1}$ at ∞ , so the sum of its finite residues is $(2g+h)/g$ and $\sum \rho_i = 1 - (2g+h)/g$. The sum of all ten exponents is then $3 = (5-2) \cdot 1$, the Fuchs relation for rank two with five singular points. $\square$

The identities of Lemma 2.1 and Theorem 2.2 are also checked symbolically [VERIFIED: $n \leq 12$, script 01_exponents.py].

Remark 2.3 (the accessory parameters and the Möbius invariants). The scaling $t \mapsto t/\mu$, i.e. $u_n \mapsto \mu^n u_n$, acts by $(a, b, c) \mapsto \mu(a, b, c)$, $(d, e, f) \mapsto \mu^2(d, e, f)$, $(g, h, j) \mapsto \mu^3(g, h, j)$, so the labelled configuration $(0, \infty; \{t_1, t_2, t_3\})$ has exactly two invariants,

$$\mathcal{I}_1 = \frac{a^3}{g} = \frac{(\lambda_1 + \lambda_2 + \lambda_3)^3}{\lambda_1 \lambda_2 \lambda_3}, \quad \mathcal{I}_2 = \frac{a^2}{d} = \frac{(\lambda_1 + \lambda_2 + \lambda_3)^2}{\sum_{i < j} \lambda_i \lambda_j},$$

the analogue of the cross-ratio invariant of [4]. Both are rational for an integral row — and, unlike the four-point case, this is no longer restrictive: the realisable locus for a five-fibre configuration is a *curve* in the $(\mathcal{I}_1, \mathcal{I}_2)$ -plane, not a point, those configurations coming in one-parameter families [14]. That is the precise sense in which the four-term case is non-rigid, and why the $\mathcal{J}$ -map test has to do all the work later.

2.2. Equal exponents.

Theorem 2.4 (normalisation classes; [PROVED]). *Assume t_1, t_2, t_3 distinct. Then $\rho_1 = \rho_2 = \rho_3 = \rho$ if and only if*

$$b = (1 - \rho)a, \quad e = -2\rho d, \quad h = -(1 + 3\rho)g.$$

Proof. $\rho_i = \rho$ for all i says $\mathcal{R} \mid T + \rho t\mathcal{R}'$. Both are cubics with constant term 1, so $T + \rho t\mathcal{R}' = \mathcal{R}$, i.e. $1 - (b + \rho a)t + (d + e + 2\rho d)t^2 - (2g + h + 3\rho g)t^3 = 1 - at + dt^2 - gt^3$. $\square$

Corollary 2.5 (Galois; [PROVED]). *The ρ_i are values at t_i of a fixed rational function with rational coefficients, hence are permuted by $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ exactly as the t_i are. Kodaira admissibility (Corollary 2.6) forces $\rho_i \in \mathbb{Q}$; therefore ρ_i is constant on each Galois orbit of $\{t_1, t_2, t_3\}$, and in particular if χ is irreducible then $\rho_1 = \rho_2 = \rho_3$.*

Corollary 2.6 (Kodaira admissibility; [PROVED] given Kodaira’s list). *If the local system of L is, up to a rational gauge factor and a quadratic twist, $R^1\pi_*\mathbb{Q}$ of an elliptic surface over $\mathbb{P}_t^1$, then $\rho_1, \rho_2, \rho_3, \delta_\infty \in \{0, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}\} \pmod{1}$.*

Proof. Neither a rational gauge nor a quadratic twist changes an exponent *difference* mod 1, and Kodaira’s monodromy eigenvalue ratios are 1 (I_n, I_n^*), -1 (III, III^*), $e^{\pm 2\pi i/3}$ (IV, IV^*), $e^{\mp 2\pi i/3}$ (II, II^*) [9, 12]. This is the argument of [4, Theorem B(ii)] verbatim. $\square$

2.3. The class parametrisation. Fix ρ and write $R(n) = C(Mn - j_1)(Mn - j_2)$, so $s_i = j_i/M$ and $g = CM^2$, $h = -CM(j_1 + j_2)$, $j = Cj_1j_2$, with $j_1 + j_2 = (1 + 3\rho)M$ by Theorems 2.2 and 2.4. A *class* is the tuple $(\rho; M, j_1, j_2)$, and inside it the scan runs over

$$P(n) = a(n^2 + (1 - \rho)n) + c, \quad Q(n) = d(n^2 - 2\rho n) + f, \quad R(n) = C(Mn - j_1)(Mn - j_2) \quad (5)$$

with the five integer parameters a, c, d, f, C (four after $t \mapsto t/\mu$). The two accessory parameters are visible as $c = u_1 = P(0)$ and $f = Q(0)$. There are 156 Kodaira-admissible classes with ρ, δ_∞ of denominator ≤ 6 , $\rho \in [-1, 2]$, $\delta_\infty \in [0, 4]$ and $M \leq 12$ [VERIFIED: 04_classes.py].

2.4. The semistable classes. “All five exponent differences $\equiv 0$ ” means $\rho \in \mathbb{Z}$ and $\delta_\infty \in \mathbb{Z}$. Since $s_1 + s_2 = 1 + 3\rho$, and $\delta_\infty = 0$ forces $s_1 = s_2 = (1 + 3\rho)/2$, the exponents at ∞ are integral iff ρ is odd. Three presentations matter:

$$\begin{aligned} (\mathbf{S}_0) \quad & P = a(n^2 + n) + c, \quad Q = dn^2 + f, \quad R = C(2n - 1)^2, \\ (\mathbf{S}_1) \quad & P = an^2 + c, \quad Q = d(n^2 - 2n) + f, \quad R = C(n - 2)^2, \\ (\mathbf{S}_{-1}) \quad & P = a(n^2 + 2n) + c, \quad Q = d(n^2 + 2n) + f, \quad R = C(n + 1)^2. \end{aligned}$$

$(\mathbf{S}_0)$ has exponents $(\frac{3}{2}, \frac{3}{2})$ at ∞ : four I_n fibres and one I_m^* . $(\mathbf{S}_{\pm 1})$ has integral exponents at ∞ and five I_n fibres; the two are gauge partners, the gauge $y \mapsto \prod_i (1 - \lambda_i t)^{-\rho} y$ sending $\rho \mapsto -\rho$ and $s \mapsto s - 3\rho$. The distinction between $\mathbf{S}_0$ and $\mathbf{S}_{\pm 1}$ is not a matter of gauge: a quadratic character twist flips the starredness of the fibres at an *even* number of points, so the parity of the number of I^* fibres is an invariant of the projective local system — 1 for $\mathbf{S}_0$, 0 for $\mathbf{S}_{\pm 1}$. This is the four-term form of the twist-parity lemma of [4].

$(\mathbf{S}_0)$ is the exact analogue of Zagier’s shape $(an^2 + an + b, cn^2)$: the linear conditions are $b = a$, $e = 0$, $h = -g$, $j = g/4$, and both accessory parameters are free — unlike the three-term case, f is *not* forced to vanish, the fibre at ∞ being governed by R , not Q . It carries integral rows; $(\mathbf{S}_{\pm 1})$ does not (Proposition 2.7).

2.5. Apparent singularities. At t_i the exponents are $(0, \rho_i)$. If $\rho_i \in \mathbb{Z} \setminus \{0\}$ the two exponents differ by an integer and the second solution either carries a logarithm (a genuine I_n fibre) or does not, in which case the local monodromy is trivial: an *apparent* singularity. The same happens at ∞ when $\delta_\infty \in \mathbb{Z}_{>0}$. If $\rho_i = 0$ (or $\delta_\infty = 0$) the exponents coincide and a logarithm is unavoidable, so the point is always a genuine I_n .

The obstruction is a single Frobenius coefficient: writing the equation at the point as $Ay'' + By' + Cy = 0$ and running the recursion from the smaller exponent, the coefficient at $m = |\rho_i|$ (resp. $m = \delta_\infty$) has vanishing indicial factor, and the point is apparent iff the accumulated right-hand side vanishes [VERIFIED: 100 digits, relative threshold 10^{-40}]. Two cases must be distinguished: with integral exponents the local monodromy is the identity and the point is not a singular point of the local system at all; with half-integral ones — which happens at ∞ when $\delta_\infty \in \mathbb{Z}$ but $s_i \notin \mathbb{Z}$ — it is $-I$, trivial in PSL_2 , i.e. an I_0^* fibre. Both are invisible to the *projective* local system, the one the $\mathcal{J}$ -map test and the Kodaira dictionary see, so in both cases the row counts as a four-point object — the I_0^* mechanism that produced the one-parameter family classes in the rigid case [4].

Proposition 2.7 (the class $\mathbf{S}_{-1}$; [PROVED]). *In the class $(\rho; \delta_\infty) = (-1; 0)$ the Frobenius obstruction at t_i is $S_i = t_i[(a - c) + (f - d)t_i]$. Hence at most one t_i is apparent unless $a = c$ and $d = f$, and in that case all three are, which happens exactly when $P(n) = a(n + 1)^2$, $Q(n) = d(n + 1)^2$, $R(n) = C(n + 1)^2$, i.e. exactly when the row degenerates to the constant-coefficient recurrence $u_{n+1} = au_n - du_{n-1} + Cu_{n-2}$ and $\sum u_n t^n$ is a rational function.*

Proof. Substitute $b = 2a$, $e = 2d$, $h = 2g$, $j = g$ into $S_i = 2A_2 - B_1 + C_0$ with $A = t^2\mathcal{R}$, $B = t\mathcal{S}$, $C = t\mathcal{V}$ expanded at t_i , and use $\mathcal{R}(t_i) = 0$ and $\mathcal{S}(t_i) = 2t_i\mathcal{R}'(t_i)$ (which is $\rho_i = -1$); this gives $S_i = t_i[2\mathcal{R}' + t_i\mathcal{R}'' - \mathcal{S}' + \mathcal{V}](t_i) = t_i[(a - c) + (f - d)t_i]$ identically in t_i . $\square$

2.6. Mixed exponents. Corollary 2.5 confines unequal exponents to reducible cubics, and the general such shape has a rational root and a quadratic Galois orbit. Let $W(\lambda) = \lambda^3 + \beta_1\lambda^2 + \beta_2\lambda + \beta_3$ be the reversal of T , i.e. $T(t) = 1 + \beta_1t + \beta_2t^2 + \beta_3t^3$.

Lemma 2.8 ([PROVED]). $W(\lambda) = \chi(\lambda)[1 + \sum_i \rho_i \lambda_i / (\lambda - \lambda_i)] = \chi(\lambda) + \sum_i \rho_i \lambda_i \prod_{j \neq i} (\lambda - \lambda_j)$, and $b = -\beta_1$, $d + e = \beta_2$, $2g + h = -\beta_3$.

Proof. With $\mathcal{R} = \prod_i (1 - \lambda_i t)$ one has $t_i \mathcal{R}'(t_i) = -(\lambda_i - \lambda_j)(\lambda_i - \lambda_k) / \lambda_i^2$, so $\rho_i = -T(t_i) / (t_i \mathcal{R}'(t_i))$ reads $W(\lambda_i) = \rho_i \lambda_i \chi'(\lambda_i)$. W is the unique *monic* cubic with those three values; Lagrange interpolation with $L_i(\lambda) = \chi(\lambda) / ((\lambda - \lambda_i) \chi'(\lambda_i))$ gives the display, the leading coefficient being 1 because the correction term has degree ≤ 2 . $\square$

Setting all $\rho_i = \rho$ recovers $W = \rho \lambda \chi' + (1 - 3\rho)\chi$, i.e. Theorem 2.4.

Theorem 2.9 (mixed normalisation forms; [PROVED]). *Let $\chi = (\lambda - r)(\lambda^2 - \sigma\lambda + \pi)$ with $r, \sigma, \pi \in \mathbb{Z}$, and put ρ_r on the rational root and ρ_p on both roots of the quadratic factor. Then $a = r + \sigma$, $d = \sigma r + \pi$, $g = r\pi$ and*

$$b = (1 - \rho_r)r + (1 - \rho_p)\sigma, \quad e = -\rho_p(2\pi + r\sigma) - \rho_r r\sigma, \quad h = -(1 + 2\rho_p + \rho_r)g,$$

and, writing $R(n) = C(Mn - j_1)(Mn - j_2)$ as in (5), $g = CM^2$, $j = Cj_1j_2$ and $(j_1 + j_2)/M = s_1 + s_2 = 1 + 2\rho_p + \rho_r$. Fuchs is automatic: $\sum \rho_i = 2\rho_p + \rho_r = s_1 + s_2 - 1$.

Proof. Expand Lemma 2.8 with $\rho_1 = \rho_r$ at $\lambda_1 = r$ and $\rho_2 = \rho_3 = \rho_p$, using $\lambda_2 \prod_{j \neq 2} (\lambda - \lambda_j) + \lambda_3 \prod_{j \neq 3} (\lambda - \lambda_j) = (\lambda - r)(\sigma\lambda - 2\pi)$, so that $W = \chi + \rho_p(\lambda - r)(\sigma\lambda - 2\pi) + \rho_r r(\lambda^2 - \sigma\lambda + \pi)$, and read off the coefficients. $\square$

Theorem 2.9 was checked numerically against the dictionary of Theorem 2.2 for all 15 shapes tried, six random parameter points each, 0 failures [VERIFIED: 01_mixcheck.py]. A mixed class is a tuple $(\rho_p, \rho_r; M, j_1, j_2)$; twenty are Kodaira-admissible, non-Casoratian and genuinely mixed:

The twenty are: $(\rho_p, \rho_r) = (\pm \frac{1}{2}, 0)$, fibres I_n, III, III , with (M, j_1, j_2) equal to $(4, 3, 5)$, $(6, 5, 7)$, $(3, 2, 4)$, $(2, 1, 3)$ resp. $(1, 0, 0)$, $(4, -1, 1)$, $(6, -1, 1)$, $(3, -1, 1)$; $(0, \pm \frac{1}{2})$, fibres III, I_n, I_n , with $(4, 3, 3)$,

$(12, 7, 11)$, $(12, 5, 13)$, $(2, 0, 3)$ resp. $(4, 1, 1)$, $(2, 0, 1)$, $(12, 1, 5)$, $(12, -1, 7)$; $(\pm\frac{1}{2}, \mp\frac{1}{2})$, three *III* fibres in two gauges, with $(4, 3, 3)$ and $(4, 1, 1)$; and the two controls $(0, \pm 1)$, where apparent points are expected.

Because $\rho_p = \pm\frac{1}{2} \notin \mathbb{Z}$ and $\rho_r = 0$ forces a logarithm, the classes in the first two rows admit *no* apparent singularity at a finite point — only ∞ can be apparent, and only when $\delta_\infty \in \mathbb{Z}$ — so they carry genuine five-point rows automatically. Structurally they are the quadratic twists, ramified at exactly two of the three finite points, of the equal-exponent classes: the twist turns two I_n fibres into I_n^* and leaves the *projective* local system unchanged. They are not a new geometry but a new *integral form* of the same geometry, and, as for the $\pm\frac{1}{2}$ twist pairs of §4.1 note 3, their Apéry limits differ from those of the untwisted rows.

2.7. The Riemann–Hurwitz window.

Theorem 2.10 ([PROVED]). *Let $\pi : \mathcal{E} \rightarrow \mathbb{P}_t^1$ be a minimal elliptic surface with section and non-constant $\mathcal{J}$ whose singular fibres lie over the five singular points of L , and let c, e_2, e_3 count the points whose projective local monodromy is parabolic, of order 2, of order 3 (so $c + e_2 + e_3 \leq 5$ and $c \geq 1$ because $t = 0$ is MUM). Then $\mu = \deg \mathcal{J} \leq 6c + 4e_3 + 3e_2 - 12 \leq 18$.*

Proof. Riemann–Hurwitz for $\mathcal{J} : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ together with Kodaira’s dictionary gives $2\mu - 2 \geq (\mu - c) + \frac{2}{3}(\mu - e_3) + \frac{1}{2}(\mu - e_2)$; maximising subject to $c + e_2 + e_3 \leq 5$ gives $\mu \leq 6 \cdot 5 - 12 = 18$, attained only in the all-parabolic case. The argument is that of the four-point bound in [4], which never used the number of singular points. $\square$

By Gauss–Bonnet a genus-zero group $\Gamma \leq \mathrm{PSL}_2(\mathbb{Z})$ with exactly five special points (c cusps, e_2 elliptic of order 2, e_3 of order 3, $c + e_2 + e_3 = 5$, $c \geq 1$) has index $\mu = 12(-1 + \frac{c}{2} + \frac{e_2}{4} + \frac{e_3}{3}) = 6c + 4e_3 + 3e_2 - 12$, which is exactly the bound of Theorem 2.10: the two agree because equality says precisely that $\mathcal{J}$ is a Belyi map. The fifteen possible signatures give indices $(5, 0, 0):18$, $(4, 1, 0):15$, $(4, 0, 1):16$, $(3, 2, 0):12$, $(3, 1, 1):13$, $(3, 0, 2):14$, $(2, 3, 0):9$, $(2, 2, 1):10$, $(2, 1, 2):11$, $(2, 0, 3):12$, $(1, 4, 0):6$, $(1, 3, 1):7$, $(1, 2, 2):8$, $(1, 1, 3):9$, $(1, 0, 4):10$, so there are *finitely many host groups* — the five-point analogue of the 28 genus-zero groups with four special points in the rigid case. Their enumeration into conjugacy classes is not carried out here [OPEN]. Accordingly the $\mathcal{J}$ -map test is run with $\mathrm{DMAX} = \mathrm{HMAX} = 18$, 130 series terms and 760 values of γ per h .

3. THE SCANS

3.1. Design. The scanner runs one class $(\rho; M, j_1, j_2)$ at a time over the five integer parameters a, c, d, f, C of (5). Integrality is tested division-free on $U_n = u_n(n!)^2$,

$$U_{n+1} = P(n)U_n - n^2Q(n)U_{n-1} + n^2(n-1)^2R(n)U_{n-2}, \quad u_n \in \mathbb{Z} \iff v_p(U_n) \geq 2v_p(n!) \ \forall p,$$

prime by prime modulo p^K with $K = 2v_p(N!) + 2$, $N = 24$. Three cheap layers precede the exact test: analytic congruences ($U_2, \dots, U_4$ — and, in the deep scans, U_5 — are *linear in C* , so $4 \mid U_2$, $36 \mid U_3$, $576 \mid U_4$, $14400 \mid U_5$ put C on an arithmetic progression), a mask-arithmetic 2-adic prefilter to $n = 14$ modulo 2^{24} , and a 64-bit prefilter for $p = 3, 5, 7, 11$ to $n = 12$. Survivors are re-verified with exact rational arithmetic to $n = 120$, which also computes the characteristic roots, the Frobenius obstruction at every point with an integral exponent difference, the companion b_n ($b_0 = 0$, $b_1 = 1$) and its sharp denominator exponent k (least k with $d_n^k b_n \in \mathbb{Z}$ for $n \leq 60$, $d_n = \mathrm{lcm}(1, \dots, n)$), the Apéry limit to ≥ 60 digits, and the score $\log(1/|\lambda_2|) - k$.

Two validations are on record: the deep scanner returns bit-identical hit lists to the first scanner on 14 classes over a common box, and the mixed-class scanner agrees with an exact brute force on a small box in the class $(-\frac{1}{2}, 0; 1, 0, 0)$, both returning the same 4 rows — soundness *and* completeness [VERIFIED: as stated].

3.2. Deduplication. A hit is discarded or reclassified by the following tests, in order.

(D1) *Repeated characteristic root*, $\Delta = 0$: two of the t_i collide.

- (D2), (D3) *Terminating rows* — three consecutive zeros kill the sequence (in the class $\mathbf{S}_1$ this alone removes all of the 6.5×10^5 raw hits at $|a| \leq 40$) — and *constant-coefficient rows*, P, Q, R all proportional to $(n+1)^2$ (Proposition 2.7).
- (D4) *Casoratian degeneracy*. For three solutions of (2) the Casoratian $W_n = \det[u_{n-2+k}^{(i)}]$ satisfies $W_{n+1} = R(n)W_n/(n+1)^2$ [PROVED], so if $R(n_0) = 0$ for an integer $n_0 \geq 1$ the solution space collapses to two dimensions from n_0 on.
- (D5) *Apparent singularity $\Rightarrow$ disguised three-term row* (§2.5): the projective local system has only four singular points, and the row is a gauge or cusp-move image of a three-term row, over $\mathbb{Q}(\lambda_i)$ in general since the Möbius map $t \mapsto t/(1 - \lambda_i t)$ is defined over that field.
- (D6) *Rescaling*: $u_n \mapsto \mu^n u_n$ acts by $(a, c, d, f, C) \mapsto (\mu a, \mu c, \mu^2 d, \mu^2 f, \mu^3 C)$; a hit is marked RESCALED when dividing by some $\mu \geq 2$ leaves an integral row.
- (D7), (D8) *Pullbacks $t \mapsto t^2$* : these would require $\mathcal{R}(-t) = \mathcal{R}(t)$, i.e. $a = g = 0$, but $g \neq 0$, so *there are none in the four-term world* [PROVED]. The Padé log/arctan families are three-term and their four-term images are removed by (D5); exactly two surviving five-point rows have $k = 1$ and are marked Padé-type by the same diagnostic.

Two mechanisms produce (D5) hits and account for essentially all of them: the *generic cusp move* $s = t/(1 - \nu t)$ with ν *not* a characteristic root, which sends the ordinary point $t = 1/\nu$ to $s = \infty$ and makes that new fifth point apparent; and a *rational gauge at an ordinary point*, $y \mapsto y/(1 - \nu t)$ with $\mathcal{R}(1/\nu) \neq 0$, which creates a fifth point with exponents $(0, -1)$ and no logarithm — and unequal ρ_i , so a reducible cubic, in accordance with Corollary 2.5.

Proposition 3.1 (the generic cusp move, four-term form; [PROVED]). *Let (P_0, Q_0) be a three-term row in Zagier’s class, $P_0 = a_0(n^2 + n) + c_0$, $Q_0 = d_0 n^2$, with integral u_n , and let $\nu \in \mathbb{Z}$ be arbitrary (in particular not a characteristic root). Then the signed binomial transform $v_n = \sum_{m=0}^n \binom{n}{m} (-\nu)^{n-m} u_m$ satisfies the four-term row of class $(\rho; M, j_1, j_2) = (0; 1, 0, 1)$ with*

$$P(n) = (a_0 - 3\nu)(n^2 + n) + (c_0 - \nu), \quad Q(n) = (d_0 - 2a_0\nu + 3\nu^2)n^2, \quad R(n) = -\nu(d_0 - a_0\nu + \nu^2)n(n-1),$$

i.e. $f = Q(0) = 0$ identically. Its characteristic roots are $\{\lambda_1 - \nu, \lambda_2 - \nu, -\nu\}$, and $t = \infty$ is an apparent singularity: the local system is the original four-point one.

Proof. $\sum_n v_n t^n = (1 + \nu t)^{-1} y(t/(1 + \nu t))$, so the singular points of the transform are the preimages $t = 1/(\lambda_i - \nu)$ of the t_i , the preimage $t = -1/\nu$ of $s = \infty$, and $t = \infty$ — the image of the *ordinary* point $s = 1/\nu$, carrying only the pole of the gauge factor, hence exponents $(0, 1)$ and no logarithm. Then $\mathcal{R}(t) = \prod_i (1 - (\lambda_i - \nu)t) \cdot (1 + \nu t)$ gives $a, d, g, c = v_1 = c_0 - \nu$, and f falls out of $4v_2 = P(1)v_1 - Q(1)$. $\square$

Proposition 3.1 was checked on Zagier **A–F** for $|\nu| \leq 3$ [VERIFIED: $n \leq 18$]. For a three-term row of another class the gauge exponent must be taken in $\{1 - r_1, 1 - r_2\}$; the resulting classes are those with $\delta_\infty \in \mathbb{Z}$, and every one is populated in the scan. This is the four-term analogue of the I_m^*/I_0^* family classes of the rigid case [4]: there the infinite families were the classical Legendre–Padé rows; here they are cusp-move images, a fifth singular point that is not really there.

3.3. The boxes. *Box G* (general census): $1 \leq a \leq 200$ (≤ 140 for the classes of the third pass), $|c| \leq 40$, $|d| \leq 1200$, $|f| \leq 80$, $1 \leq |C| \leq 120$, integrality prime-tested to $n = 24$ and re-verified exactly to $n = 120$.

Box W ($|\lambda_2| < 1$; complete). A positive score needs $|\lambda_2| < 1$, hence $|\lambda_3| \leq |\lambda_2| < 1$, and from $a = \sum \lambda_i$, $d = \sum_{i < j} \lambda_i \lambda_j$, $g = \lambda_1 \lambda_2 \lambda_3$ this forces [PROVED]

$$|\lambda_1| \leq |a| + 2, \quad |d| \leq 2|a| + 5, \quad |C| = |g|/M^2 \leq (|a| + 2)/M^2.$$

Box W takes $|c| \leq 60$, $|f| \leq 120$ and lets $|d|, |C|$ run over exactly those ranges; swept to $a \leq 300$ for 18 classes and $a \leq 400$ for $\mathbf{S}_0$, it is *complete* for the question “is there a row with $|\lambda_2| < 1$?”.

Box PW (positive-score window; complete). Every census row has $k \geq 1$, so a positive score needs $|\lambda_2| < e^{-1}$, whence [PROVED]

$$|\lambda_1| \leq |a| + 2e^{-1}, \quad |d| \leq 0.74(|a| + 1) + 2, \quad |C| \leq (0.136(|a| + 1) + 1)/M^2.$$

Box PW is defined by $1 \leq a \leq 1500$, $|c| \leq 60$, $|f| \leq 120$ over all 27 classes and was swept to completion for $a \leq 800$; inside it the scan is complete for the question “is there a row with positive score?” among rows with $k \geq 1$.

The deep boxes. A later run added box DG (the 15 carrier classes, $|c| \leq 200$, $|d| \leq 2000$, $|f| \leq 400$, $|C| \leq 150$), box DW (all 27 equal-exponent classes, $|c| \leq 250$, $|f| \leq 500$, complete in (d, C) for every $a \leq 180$, about $17\times$ the (c, f) volume of box W), box MIX (the 20 mixed classes, $a \leq 400$, $|c| \leq 100$, $|d| \leq 3000$, $|f| \leq 200$, $|C| \leq 400$, sharded by the rational characteristic root r) and box F5 (seven five-term classes, §6). The cost is proportional to the product of the five ranges, so the boxes were sized to budget with the accessory directions $c = u_1$ and $f = Q(0)$ — where the period lives — given priority over a and C .

3.4. What the scans produce. Raw hit counts are dominated by degenerate and disguised rows. Two classes make the point on their own: in $\mathbf{S}_1$ the box $a \leq 40$, $|c| \leq 30$, $|d| \leq 600$, $|f| \leq 50$, $|C| \leq 80$ contains 646 297 parameter tuples passing integrality mod $p \leq 24$ to $n = 24$ with $\Delta \neq 0$, and every one of them has $u_n = 0$ for $n \geq 1$; in $\mathbf{S}_{-1}$ the same box contains 484 697, and every one of them is the constant-coefficient row of Proposition 2.7. After (D2)–(D3) both classes are empty.

Over the whole census — 27 classes, box G for 21 of them and a probe box for the six purely disguised ones — the filters act as follows [VERIFIED: boxes as stated]:

stage	rows
integral mod $p \leq 24$ to $n = 24$, $\Delta \neq 0$, non-terminating, not constant-coefficient	2178
of which exact integrality to $n = 120$ fails	4
of which (D5) an apparent singularity: four-point local system, a disguised three-term row	2138 (98.2%)
of which (D6) a rescaling $u_n \mapsto \mu^n u_n$ of another integral row	6
genuine five-point rows, primitive	30

Two observations are stable across all boxes. First, the classes producing 10^5 – 10^6 raw hits are exactly the Casoratian-degenerate ones; in each the hits are terminating or constant-coefficient, so after (D2)–(D3) the class is empty, the one survivor being $(0; 1, 0, 1)$, $R = Cn(n-1)$ — the class of Proposition 3.1 — all of whose rows are disguised. Second, every class with $\rho \in \mathbb{Z} \setminus \{0\}$ or $\delta_\infty \in \mathbb{Z}_{>0}$ produces a large family with vanishing Frobenius obstruction throughout, the extreme case being $(1; 2, 3, 5)$ with *two* apparent points, whose rows are hypergeometric. The classes with $\rho \notin \mathbb{Z}$, or with $\rho = 0$ and $\delta_\infty \notin \mathbb{Z}$, admit no apparent singularity at all, and they carry the census.

The later deep sweep confirms the picture and adds nothing to it: over boxes DG and DW (DG $a \leq 12$, DW complete to $a \leq 180$) there are 251 distinct integral rows, 232 (92.4%) disguised, 1 rescaled and 18 candidates — *all 18 already in the list of 30*, so zero new equal-exponent five-point rows in the region covered [VERIFIED: as stated]. The disguise fraction is 92% rather than 98% only because the deep box is weighted towards the carrier classes.

4. THE SPORADIC ROWS AND THEIR PERIODS

4.1. The list of thirty. Table 1 lists all 30 primitive rows with five genuine singular points produced by boxes G and W, sorted by score. $(\rho; M, j_1, j_2)$ is the class, (a, c, d, f, C) the parameters of (5), λ_i the characteristic roots, k the sharp denominator exponent of the companion, $\text{score} = \log(1/|\lambda_2|) - k$, and $\mathcal{J}$ the outcome of the $\deg \leq 18$ $\mathcal{J}$ -map test.

Four notes.

1. $k = 2$ for 28 of the 30 — the four-term echo of the free integration phenomenon of the rigid case. The two exceptions have $k = 1$, lie in the classes with $\delta_\infty > 1$, and are Padé constructions rather than modular objects by the $k = 1$ diagnostic.
2. Exactly one row is an elliptic surface: the $K3$ of §4.4. The other 29 fail the $\mathcal{J}$ -map test over the whole $\deg \leq 18$ window — the four-term half of the non-congruence picture. A *positive* $\mathcal{J}$ answer is a certificate; a *negative* one means only “no fit with $h \leq 18$, $\deg \leq 18$ and γ in the 760-element list” [VERIFIED: as stated].

TABLE 1. The 30 genuine five-singular-point rows. [VERIFIED: integrality to $n = 120$; k to $n = 60$; obstructions at 100 digits]

class	(a, c, d, f, C)	$\lambda_1, \lambda_2, \lambda_3$	k	score	$\mathcal{I}_1$	$\mathcal{I}_2$	$\mathcal{J}$
(0; 2, 1, 1)	(154, 42, -128, -12, 8)	$154.8, -0.414 \pm 0.1878i$	2	-1.2117	456533/4	-5929/32	no
(0; 2, 1, 1)	(87, 24, -152, -16, 16)	$88.72, -1, -0.7214$	2	-2.0000	658503/64	-7569/152	no
$(\frac{1}{2}; 4, 5, 5)$	(72, 6, -192, -60, 8)	$74.6, -1.298 \mp 0.1732i$	2	-2.2700	2916	-27	no
(0; 3, -1, 4)	(22, 6, -72, 12, 36)	$25.35, -1.673 \pm 3.16i$	1	-2.2741	2662/81	-121/18	no
(0; 2, 1, 1)	(22, 6, -96, -12, 24)	$25.86, -2, -1.856$	2	-2.6931	1331/12	-121/24	no
(0; 2, 1, 1)	(11, 3, -73, -8, 28)	$16, -2.5 \pm 0.866i$	2	-2.9730	1331/112	-121/73	no
$(-\frac{1}{2}; 4, -1, -1)$	(16, 10, 64, 20, 4)	$10.47, 4, 1.528$	2	-3.3863	64	4	no
$(\frac{1}{2}; 4, 5, 5)$	(16, 2, 64, 20, 4)	$10.47, 4, 1.528$	2	-3.3863	64	4	no
(0; 2, 1, 1)	(60, 18, 240, 12, 16)	$55.71, 4, 0.2872$	2	-3.3863	3375	15	no
(0; 2, 1, 1)	(20, 6, -80, -4, -16)	$23.31, -4, 0.6863$	2	-3.3863	-125	-5	no
(0; 2, 1, 1)	(3, 0, -104, -16, 80)	$12.94, -5, -4.944$	2	-3.6094	27/320	-9/104	no
(0; 3, 1, 2)	(11, 4, 37, 3, 3)	$5 \pm 1.414i, 1$	2	-3.6479	1331/27	121/37	yes, (4, 16)
(0; 2, 1, 1)	(16, 6, 96, 12, 64)	$8, 4 \pm 4i$	2	-3.7329	16	8/3	no
(0; 2, 1, 1)	(16, 6, 76, 8, 24)	$8, 6, 2$	2	-3.7918	128/3	64/19	no
(0; 2, 1, 1)	(16, 6, 96, 12, 48)	$6 \pm 3.464i, 4$	2	-3.9356	64/3	8/3	no
(0; 4, 1, 3)	(14, 5, 97, 8, 12)	$5.5 \pm 5.809i, 3$	2	-4.0794	343/24	196/97	no
(0; 2, 1, 1)	(16, 6, 16, -4, -96)	$12, 8, -4$	2	-4.0794	-32/3	16	no
(0; 6, -1, 7)	(13, 4, 432, -24, -48)	$8.262 \pm 20.54i, -3.525$	1	-4.0975	-2197/1728	169/432	no
(0; 2, 1, 1)	(6, 2, 64, 4, -40)	$4 \pm 8i, -2$	2	-4.1910	-27/20	9/16	no
(0; 3, 1, 2)	(3, 0, -216, -24, 108)	$18, -9, -6$	2	-4.1972	1/36	-1/24	no
(0; 3, 1, 2)	(33, 12, 324, 36, 108)	$18, 9, 6$	2	-4.1972	1331/36	121/36	no
(0; 3, -1, 4)	(3, 0, -216, 48, 108)	$18, -9, -6$	2	-4.1972	1/36	-1/24	no
(0; 2, 1, 1)	(17, 6, 112, 8, 24)	$8 \pm 5.657i, 1$	2	-4.2822	4913/96	289/112	no
(0; 4, 1, 3)	(28, 10, 208, 12, 16)	$16, 10.47, 1.528$	2	-4.3487	343/4	49/13	no
(0; 6, 1, 5)	(17, 6, 56, 0, -12)	$10.21 \pm 4.658i, -3.428$	2	-4.4182	-4913/432	289/56	no
(0; 2, 1, 1)	(32, 12, 272, 16, 64)	$16, 14.93, 1.072$	2	-4.7033	128	64/17	no
(0; 2, 1, 1)	(42, 15, 441, 24, 100)	$25, 16, 1$	2	-4.7726	9261/50	4	no
$(-\frac{1}{3}; 1, 0, 0)$	(42, 22, 480, 64, -64)	$21.07 \pm 6.464i, -0.1318$	2	-5.0926	-9261/8	147/40	no
$(\frac{1}{3}; 1, 1, 1)$	(42, 8, 480, 64, -64)	$21.07 \pm 6.464i, -0.1318$	2	-5.0926	-9261/8	147/40	no
(0; 2, 1, 1)	(64, 24, 1030, 26, 48)	$32, 31.81, 0.1886$	2	-5.4598	4096/3	2048/515	no

3. *Quadratic-twist pairs.* (16, 10, 64, 20, 4) and (16, 2, 64, 20, 4), and likewise (42, 22, 480, 64, -64) and (42, 8, 480, 64, -64), have the same (a, d, f, C) and differ only in the accessory c , hence have the same five singular points; in each pair the classes are exchanged by the twist $\prod_i (1 - \lambda_i t)^{\mp 1/2}$, so they are the two half-integral gauges of one projective local system. *Their Apéry limits differ.*
4. The best score is -1.21, attained by (154, 42, -128, -12, 8) in $\mathbf{S}_0$, whose two small roots are the complex pair $-0.4140 \pm 0.1878i$; its Apéry limit $\xi = 0.02358580301301982283 \dots$ is unidentified. Adding a fifth *genuine* singular point to a known four-point configuration is a recurring motif: (87, 24, -152, -16, 16) has $\chi = (\lambda + 1)(\lambda^2 - 88\lambda - 64)$, the quadratic being the level-5 Fricke row's, and (20, 6, -80, -4, -16) has $(\lambda + 4)(\lambda^2 - 24\lambda + 16)$, the quadratic being Beukers' row's.

4.2. The periods of the thirty: a negative result. For each row the companion b_n satisfies $LB = t$, and $\xi = \lim b_n/a_n$ exists exactly when the largest characteristic root is real and simple, with error $O((\lambda_2/\lambda_1)^n)$. Nine of the thirty have a complex conjugate dominant pair and therefore *no archimedean Apéry limit* — in particular the $K3$ row, whose dominant roots are $5 \pm i\sqrt{2}$.

The 21 limits that do exist were computed to ≥ 60 digits and put through two batteries: an integer-relation search over 103 classical weight-two constants ($\zeta(2)$; $L(2, \chi_D)$ for $|D| \leq 24$; $\log^2 m$, $\pi \log m$ and their quadratic-unit variants; $\pi \arctan \frac{1}{m}$, $\arctan^2 \frac{1}{m}$; $\Gamma(1/4)^4/\pi$, $\Gamma(1/3)^6/\pi^2$; Li_2 at twenty small rational arguments), and, for each row separately, the weight-two polylogarithm space attached to *its own* singular points, built from the ratios λ_i/λ_j and $1 - \lambda_i/\lambda_j$ and reduced to a greedy independent subset first (without that reduction the classical dilogarithm identities make the basis dependent and the search returns them instead of anything about ξ).

All 21 limits came back unidentified, in both batteries, at 60 digits with coefficient heights up to 10^6 (single constant) and 10^5 (pairs), and in polylogarithm spaces of dimension 7–11 [VERIFIED: stated bases

and heights] — a negative result, but sharp, and consistent with the geometry: a five-point configuration is not a modular curve.

4.3. The complex fold constants. For the nine rows with no archimedean limit the natural invariants are the *fold constants*: continue (A, A') and (B, B') from inside $|t| < \min |t_i|$ to a singular point t_i , match to the local Frobenius basis (logarithmic for $\rho \in \mathbb{Z}$, $\{1 + \dots, s^\rho(1 + \dots)\}$ for $\rho \notin \mathbb{Z}$), and set

$$\xi(t_i) = \frac{\text{coefficient of the singular basis element in } B}{\text{coefficient of the singular basis element in } A}.$$

The machinery self-tests on the $K3$ row, reproducing its three published constants to 210–211 digits, and its non-integer- ρ branch was validated on the two $\rho = \pm \frac{1}{3}$ rows. All nine rows succeed: 27 fold constants, each self-agreeing to 154–156 digits between working precisions 130 and 210 [VERIFIED: as stated]. In each of the nine the three finite singular points are one real point and a conjugate pair, so each row contributes three real periods $\xi(t_1)$, $\xi_+ + \xi_-$, $(\xi_+ - \xi_-)/i$; and $\xi(t_1) = \xi_+ + \xi_-$ only for the $K3$ row, so the other eight contribute two independent real periods each.

Put through the 145-constant battery of §4.9, the only hits are the $K3$ row's, returning the expected $L(g, 2)$ multiples at 124–125 digits — a strong positive control. *Every one of the 24 genuinely new real periods is unidentified* [VERIFIED: as stated].

One cross-row coincidence stands out: $\text{Re } \xi(t_2) = 0.14639462825244069157\dots$ of the row $(0; 4, 1, 3)$, $(14, 5, 97, 8, 12)$ equals the archimedean Apéry limit of the *different* row $(0; 2, 1, 1)$, $(42, 15, 441, 24, 100)$ to all 70 stored digits. The two operators have different singular points — $\lambda = \{3, \frac{11 \pm i\sqrt{115}}{2}\}$ against $\{1, 16, 25\}$ — so this is neither a twist nor a rescaling [NUMERICAL: 70 digits]. Unexplained; see Conjecture 6.2.

4.4. The elliptic $K3$ row. The row

$$(n+1)^2 u_{n+1} = (11n^2 + 11n + 4)u_n - (37n^2 + 3)u_{n-1} + 3(3n-1)(3n-2)u_{n-2}, \quad (6)$$

$u_n = 1, 4, 16, 64, 250, 928, 3136, 8704, 11866, -79400, \dots$, lies in the class $(0; 3, 1, 2)$, i.e. $(\rho; \delta_\infty) = (0; \frac{1}{3})$: exponents $(0, 0)$ at $t = 0$ and at each t_i , and $(\frac{5}{3}, \frac{4}{3})$ at ∞ . Its characteristic polynomial is $\lambda^3 - 11\lambda^2 + 37\lambda - 27 = (\lambda - 1)(\lambda^2 - 10\lambda + 27)$, so the singular points are $0, 1, \frac{5 \pm i\sqrt{2}}{27}, \infty$; integrality is verified exactly to $n = 300$ (u_{300} has 213 decimal digits) and $k = 2$ [VERIFIED: as stated]. The two dominant characteristic roots are the conjugate pair $5 \pm i\sqrt{2}$, so there is no archimedean Apéry limit.

The $\mathcal{J}$ -map test is positive, returning $h = 4$, $\deg \mathcal{J} = 16$, $\gamma = -1$, certified: $\mathcal{J} = U/V$ with $\deg U = 15$, $\deg V = 16$ fitted from 38 coefficients of the q -expansion and verified on 73 further ones [VERIFIED: as stated]. Factoring V , U and $U - 1728V$ gives the fibres I_4 at $t = 0$, I_6 at $t = 1$, I_3, I_3 at $t = \frac{5 \pm i\sqrt{2}}{27}$ and IV^* at ∞ , with $\mathcal{J}^{-1}(\infty) : 4 + 6 + 3 + 3 = 16$, $\mathcal{J}^{-1}(0) : 3 \cdot 5 + 1 = 16$, $\mathcal{J}^{-1}(1728) : 2 \cdot 8 = 16$ and $\sum_P (e_P - 1) = 30 = 2 \cdot 16 - 2$: $\mathcal{J}$ is a *Belyi map*, so there are no further singular fibres, and the projective monodromy is the genus-zero group of signature $(0; 3; \text{cusps } 3, 3, 4, 6)$ and index 16 — the $(c, e_2, e_3) = (4, 0, 1)$ line of §2.7, at the Riemann–Hurwitz bound $6 \cdot 4 + 4 \cdot 1 - 12 = 16$ with equality.

Since $\sum_{I_n} n_i = 16 > 12$ the surface is not rational; the Euler numbers give $\sum e = 4 + 6 + 3 + 3 + 8 = 24$, so $\chi = 2$: an *elliptic $K3$ surface over $\mathbb{P}_t^1$ with exactly five singular fibres $I_4 I_6 I_3 I_3 IV^*$* , defined over $\mathbb{Q}$ with the two I_3 's conjugate over $\mathbb{Q}(\sqrt{-2})$.

Theorem 4.1 ([5, 15]; stated here, not reproved). *For the row (6): the local system is $R^1 \pi_* \mathbb{Q}$ of an elliptic $K3$ surface with the fibres above, given by an explicit Weierstrass model [PROVED]; $\rho(\mathcal{X}) = 20$, MW has rank 0 and torsion exactly $\mathbb{Z}/3$ [PROVED]; the transcendental lattice is $\text{diag}(6, 12)$, $\det T = 72$, with CM by $\mathbb{Q}(\sqrt{-2})$ [PROVED] modulo the discriminant-form bookkeeping; the transcendental motive is that of the weight-3 CM newform $g = 32.3.d.a \in S_3(\Gamma_0(32), \chi_{-8})$ [VERIFIED: a_p for all 76 primes $5 \leq p \leq 397$, no free parameter, Livné rigidity then identifying the motive [10]]; and the fold constants are*

$$\xi\left(\frac{5 \pm i\sqrt{2}}{27}\right) = \frac{\sqrt{2} \mp i}{3} L(g, 2), \quad \xi(1) = \frac{2\sqrt{2}}{3} L(g, 2) = \xi_+ + \xi_-, \quad L(g, 2) = \frac{\pi}{32} \frac{\Gamma(\frac{1}{8})\Gamma(\frac{3}{8})}{\Gamma(\frac{5}{8})\Gamma(\frac{7}{8})}$$

[VERIFIED: ≥ 165 digits, two independent computations]. 2-adically $b_n/a_n \rightarrow 0$ at logarithmic rate and there is no 3-adic limit [VERIFIED: as stated in [15]].

This is the structural pay-off of the fifth point: *the four-term row is the recurrence-level home of the elliptic K3s with five singular fibres, just as the three-term row is the home of the rational elliptic surfaces with four*. The integrality of (6) is nonetheless purely experimental [OPEN]— for three-term rows integrality is explained by modular parametrisation, and here we do not know whether the index-16 monodromy group is congruence.

4.5. The mixed-exponent rows: Catalan and L -values. The mixed classes of §2.6 were scanned next, and they carry everything the equal-exponent scan does not. Six genuine five-point rows with identified classical periods appear, all in the single class

$$(\rho_p, \rho_r; M, j_1, j_2) = (-\tfrac{1}{2}, 0; 1, 0, 0), \quad R(n) = Cn^2,$$

i.e. exponent differences $(0, \frac{1}{2}, \frac{1}{2})$ at the three finite points and $\delta_\infty = 0$: one I_n fibre and two points whose projective monodromy has order 2 (Kodaira III/III^*), with I_n 's at 0 and ∞ .

TABLE 2. The six mixed-exponent rows with classical Apéry limits. [VERIFIED: integrality to $n = 200$; exponents to 40 digits; k to $n = 80$; ξ to 139–141 digits]

#	$P(n)$	$Q(n)$	$R(n)$	u_n	λ	score	$\xi = \lim b_n/a_n$
1	$16n^2+20n+8$	$48n^2+16n$	$-128n^2$	$1, 8, 72, 640, \dots$	$4\pm 4\sqrt{2}, 8$	-4.0794	$\frac{1}{4}G$
2	$14n^2+20n+8$	$28n^2+16n+4$	$8n^2$	$1, 8, 72, 704, \dots$	$6\pm 4\sqrt{2}, 2$	-2.6931	$\frac{1}{2}G - \frac{3}{16}\zeta(2)$
3	$6n^2+10n+4$	$-32n^2-24n-8$	$32n^2$	$1, 4, 36, 288, \dots$	$4\pm 4\sqrt{2}, -2$	-2.6931	$\frac{3}{8}\zeta(2) - \frac{1}{2}G$
4	$16n^2+20n+8$	$68n^2+36n+8$	$32n^2$	$1, 8, 60, 448, \dots$	$8, 4\pm 2\sqrt{3}$	-4.0101	$\frac{1}{32}(2\zeta(2) + 15L_{-3})$
5	$17n^2+25n+10$	$32n^2+24n+8$	$16n^2$	$1, 10, 114, 1424, \dots$	$8\pm 4\sqrt{3}, 1$	-2.0693	$\frac{1}{16}(15L_{-3} - 6\zeta(2))$
6	$13n^2+20n+8$	$-13n^2-6n-1$	$-n^2$	$1, 8, 87, 1024, \dots$	$7\pm 4\sqrt{3}, -1$	-2.0000	$\frac{1}{8}(2\zeta(2) - 3L_{-3})$

Here $G = L(2, \chi_{-4})$, $L_{-3} = L(2, \chi_{-3})$, all six have $k = 2$, and the class parameters $(a, c, d, f, C)_r$ are $(16, 8, 48, 0, -128)_8$, $(14, 8, 28, 4, 8)_2$, $(6, 4, -32, -8, 32)_{-2}$, $(16, 8, 68, 8, 32)_8$, $(17, 10, 32, 8, 16)_1$, $(13, 8, -13, -1, -1)_{-1}$. Independently of the scan pipeline the coefficients were rebuilt from Theorem 2.9 and, for each row: $u_n \in \mathbb{Z}$ by exact rational arithmetic to $n = 200$; the exponents recomputed from the dictionary come out as $(0, -\frac{1}{2}, -\frac{1}{2})$ to 40 digits with cubic discriminants 2048, 32768, 32768, 768, 192, 49152, all non-zero, so the five singular points are distinct; *no apparent singularity is possible*, since $\rho_r = 0$ and $\delta_\infty = 0$ force logarithms and the two III points have monodromy of order 2 in PSL_2 ; $k = 2$ to $n = 80$; ξ recomputed by forward recursion at working precision 380 agrees to 139–141 digits; and *the sequences satisfy no three-term recurrence* — the nine-parameter ansatz $L(n)u_{n+1} = P(n)u_n - Q(n)u_{n-1}$ with L, P, Q quadratic has nullspace dimension 0 over 21 equations, while the twelve-parameter four-term ansatz has nullspace dimension exactly 1. [VERIFIED: as stated]

The finite singular points of rows 1–3 are irrational and conjugate over $\mathbb{Q}(\sqrt{2})$, the field of the level-8 and level-16 Catalan hosts; rows 4–6 are the same phenomenon over $\mathbb{Q}(\sqrt{3})$. Rows 2 and 3 realise the mixed class $\frac{1}{2}G \mp \frac{3}{16}\zeta(2)$, $\frac{3}{8}\zeta(2) - \frac{1}{2}G$ on a five-point host. The best Catalan score is $-2.6931 = \log \frac{1}{2} - 2$, i.e. $+0.693$ nats better than Zagier $\mathbf{E}$'s -3.3863 , and row 6 reaches -2.0000 with $|\lambda_2| = 1$ exactly. For rows 1, 2, 3, 5, 6 the singular point nearest the origin — the one that governs $\lim b_n/a_n$ — is one of the two III points, an order-2 orbifold point rather than a cusp; for row 4 it is the cusp $t = \frac{1}{8}$. Every one of the six has $|\lambda_2| \geq 1$, so none is a decayer, as Theorem 5.1 forces.

4.6. The whole period matrix is classical. The Apéry limit is only the fold constant at the nearest singular point. The constants at all three were computed and identified, real and imaginary parts, at 125–129 digits [VERIFIED: as stated]. Ordering t_1, t_2, t_3 by decreasing $|t|$, so that $\xi(t_3)$ is the archimedean Apéry limit:

row	$\xi(t_1)$	$\xi(t_2)$	$\xi(t_3)$
$(16, 8, 48, 0, -128)_8$	$\frac{1}{8}(2G - 3\zeta(2))$	$\frac{1}{32}(8G + 3\zeta(2)) + \frac{3}{32}\zeta(2)i$	$\frac{1}{4}G$
$(14, 8, 28, 4, 8)_2$	$\frac{1}{80}(40G - 39\zeta(2)) + \frac{9}{10}\zeta(2)i$	$\frac{1}{16}(8G + 3\zeta(2)) + \frac{1}{8}\zeta(2)i$	$\frac{1}{16}(8G - 3\zeta(2))$
$(16, 8, 68, 8, 32)_8$	$\frac{1}{32}(15L_{-3} - 12\zeta(2)) + \frac{5}{96}\pi^2\sqrt{3}i$	$\frac{1}{32}L_{-3} + \frac{1}{96}\pi^2\sqrt{3}i$	$\frac{1}{32}(2\zeta(2) + 15L_{-3})$
$(8, 4, 32, 8, 64)_4$	$\frac{1}{32}(6\zeta(2) + 15L_{-3})$	$\frac{1}{32}(15L_{-3} - 6\zeta(2)) + \frac{1}{48}\pi^2\sqrt{3}i$	$\xi(t_2)$

Two consequences. First, the period lattice is spanned by two classical constants: for the $\mathbb{Q}(\sqrt{2})$ family every fold period lies in $\mathbb{Q}G + \mathbb{Q}\zeta(2)$, for the $\mathbb{Q}(\sqrt{3})$ family in $\mathbb{Q}L_{-3} + \mathbb{Q}(\sqrt{3})\zeta(2)$ (note $\pi^2\sqrt{3} = 6\sqrt{3}\zeta(2)$). These are *not* generic five-point periods — all 21 equal-exponent Apéry limits and all 24 new equal-exponent fold periods are unidentified over the same battery — so the mixed class is arithmetically special. Second, the row $(8, 4, 32, 8, 64)_4$ has no archimedean Apéry limit at all, its dominant roots being the complex pair $2 \pm 2\sqrt{3}i$, and yet all three of its fold periods are classical: the phenomenon belongs to the local system, not to the recurrence's convergence.

4.7. The gauge partners and $\Gamma(\frac{1}{4})^4$. The class $(-\frac{1}{2}, 0; 1, 0, 0)$ has the gauge partner $(+\frac{1}{2}, 0; 1, 1, 1)$ — the same projective local system with $\rho \mapsto -\rho$, the $\pm\frac{1}{2}$ twist pairing of §4.1 note 3. It has $R(n) = C(n-1)^2$, so $R(1) = 0$: a Casoratian degeneracy in the sense of (D4), which is why it was initially left out of the class list. Its integral rows have the *same* (a, d, f, C) as the Catalan rows with a different accessory c :

$$\begin{aligned} (16, 4, 48, 0, -128)_8 &: 1, 4, 24, 160, 1136, 8384, \dots \\ (14, 2, 28, 4, 8)_2 &: 1, 2, 8, 48, 356, 2952, \dots \\ (6, 0, -32, -8, 32)_{-2} &: 1, 0, 4, 16, 116, 800, \dots \end{aligned}$$

and *all three* have the same Apéry limit,

$$\xi = \frac{\Gamma(\frac{1}{4})^4}{64\pi} = 0.85939822725254660343626197247631964973760705647739511203431098644200041 \dots,$$

[VERIFIED: integrality to $n = 200$; ξ matched to 195, 241, 241 digits]. Three different rows, three different singular configurations, one and the same period — in sharp contrast to the $-\frac{1}{2}$ gauge, where the three partners give three different constants.

The pairing is the point: $\Gamma(\frac{1}{4})^4/\pi$ is the period of the CM elliptic curve $y^2 = x^3 - x$ — the algebraic-period side of the Gaussian world — and $G = L(2, \chi_{-4})$ is its L -value side, and the two half-integral gauges of one five-point projective local system over $\mathbb{Q}(\sqrt{2})$ produce one each; $\sqrt{2}$ is the ramification of $\mathbb{Q}(i)/\mathbb{Q}$ made visible in the singular points.

4.8. The complete census of the two productive classes, and ξ^* . The pair of gauge-partner classes was swept to completion over

$$|r| \leq 60, \quad |a| \leq 200, \quad |c| \leq 200, \quad |d| \leq 2000, \quad |f| \leq 400, \quad |C| \leq 600$$

(69 distinct integral rows, of which 30 rescaled, 14 not integral at $n = 120$, and 20 candidates). *Each class contains exactly nine primitive genuine five-point rows*, and they correspond one-to-one under the twist $\rho \mapsto -\rho$ — same (a, d, f, C) and same r , different accessory c [VERIFIED: box as stated]:

r	(a, d, f, C)	λ	score	ξ in gauge $\rho_p = -\frac{1}{2}$	ξ in gauge $\rho_p = +\frac{1}{2}$
8	$(16, 48, 0, -128)$	$4 \pm 4\sqrt{2}, 8$	-4.0794	$\frac{1}{4}G$	$\Gamma(\frac{1}{4})^4/(64\pi)$
2	$(14, 28, 4, 8)$	$6 \pm 4\sqrt{2}, 2$	-2.6931	$\frac{1}{2}G - \frac{3}{16}\zeta(2)$	$\Gamma(\frac{1}{4})^4/(64\pi)$
-2	$(6, -32, -8, 32)$	$4 \pm 4\sqrt{2}, -2$	-2.6931	$\frac{3}{8}\zeta(2) - \frac{1}{2}G$	$\Gamma(\frac{1}{4})^4/(64\pi)$
8	$(16, 68, 8, 32)$	$8, 4 \pm 2\sqrt{3}$	-4.0101	$\frac{1}{32}(2\zeta(2) + 15L_{-3})$	ξ^*
1	$(17, 32, 8, 16)$	$8 \pm 4\sqrt{3}, 1$	-2.0693	$\frac{1}{16}(15L_{-3} - 6\zeta(2))$	ξ^*
-1	$(13, -13, -1, -1)$	$7 \pm 4\sqrt{3}, -1$	-2.0000	$\frac{1}{8}(2\zeta(2) - 3L_{-3})$	ξ^*
4	$(8, 32, 8, 64)$	$4, 2 \pm 2\sqrt{3}i$	-3.3863	complex pair, no limit	complex pair, no limit
3	$(17, 123, 33, 243)$	$7 \pm 4\sqrt{2}i, 3$	-4.1972	complex pair, no limit	complex pair, no limit
-3	$(5, 24, 12, -144)$	$4 \pm 4\sqrt{2}i, -3$	-3.9356	complex pair, no limit	complex pair, no limit

Every row with an archimedean limit in the $-\frac{1}{2}$ gauge is identified; in the $+\frac{1}{2}$ gauge three give $\Gamma(\frac{1}{4})^4/(64\pi)$ and three the same *unidentified* constant

$$\xi^* = 0.7372929961855962401764261978022936979609191432765291852228688851129961062186382\dots$$

shared by those three rows, by a fourth four-term row in a *different* mixed class $((0, \frac{1}{2}; 2, 1, 2), r = 12, (a, c, d, f, C) = (24, 6, 192, 24, 144))$, and by the five-term six-point row of §6 whose characteristic quartic $(\lambda - 4)(\lambda - 2)(\lambda + 2)(\lambda - 1)$ has all roots rational — two computations with recurrences of different orders and different singular sets, agreeing to 319 digits [VERIFIED: 319 agreeing digits]. ξ^* is unidentified by the 145-constant battery (heights 10^8 for single constants, 10^5 for pairs, `algdep` to degree 6), and a hand search against $\Gamma(\frac{1}{3})^6/\pi^2$, $\Gamma(\frac{1}{4})^4/\pi$ and rational multiples returns only height- 10^{55} – 10^{85} noise. Five rows, in three classes and two worlds, over two fields, with one common transcendental: identifying ξ^* is the sharpest single question this work leaves behind [OPEN].

4.9. The battery. The identifications above and the negative results of §4.2 and §4.3 use a Catalan-focused battery of 145 constants after value-deduplication: G and six algebraic multiples of it; $\zeta(2)$, π^2 , $\pi^2\sqrt{D}^{\pm 1}$ for $D \mid 24$; $L(2, \chi_D)$ for all 17 fundamental discriminants $|D| \leq 24$; $L(3, \chi_D)$ for $|D| \leq 8$, $\zeta(3)$ and $\zeta(3)/\pi^{1,2}$; Ti_2 at 11 arguments; the $K3$ row's $L(g, 2)$ and four twists; seven Γ/CM -period products of levels 3, 4, 6, 8, 12; and $\log^2 m$, $\pi \log m$, 28 products $\log m_1 \log m_2$, five real-quadratic-unit pairs, $\pi \arctan \frac{1}{m}$ and $\arctan^2 \frac{1}{m}$. Constants are built at 250 digits, the search is run at 60, and *every* candidate relation is verified by reconstructing ξ from it with a 55-digit threshold; the pair test uses a 24-element sublist greedily reduced to an independent basis of 22. The battery recovers $\frac{7}{3}G$, $\frac{5}{2} - \frac{3}{7}\pi G$, $\frac{2}{5}G - \frac{3}{4}\zeta(2) + \frac{1}{6}$ (also from a 62-digit truncated input), $\sqrt{5}$, $\frac{2\sqrt{2}}{3}L(g, 2)$ and $\frac{1}{2}G$ from Zagier **E**: it is not silently blind [VERIFIED: as stated].

5. STRUCTURE THEOREMS

The census results above are computations over boxes. Three statements explain, rather than merely record, part of what they show.

Theorem 5.1 ([PROVED]). *Let (2) be an integral four-term row with $g = CM^2 \neq 0$ and characteristic roots $|\lambda_1| \geq |\lambda_2| \geq |\lambda_3|$. If $|\lambda_2| < 1$ then $\chi(\lambda) = \lambda^3 - a\lambda^2 + d\lambda - g$ is irreducible over $\mathbb{Q}$.*

Proof. Suppose χ has a rational root. It is monic with integer coefficients, so that root is an integer r , and $r \neq 0$ because $\chi(0) = -g \neq 0$. If $r \in \{\lambda_2, \lambda_3\}$ then $|r| \leq |\lambda_2| < 1$ with r a non-zero integer, which is impossible. Hence $r = \lambda_1$ and, by Gauss's lemma, $\chi = (\lambda - r)(\lambda^2 - \sigma\lambda + \pi)$ with $\sigma, \pi \in \mathbb{Z}$. Then $\pi = \lambda_2\lambda_3$ and $|\pi| \leq |\lambda_2|^2 < 1$, so $\pi = 0$ and $g = r\pi = 0$ — a contradiction. $\square$

Corollary 5.2 ([PROVED]). *No class with $\rho_1 \neq \rho_2 = \rho_3$ — in particular none of the twenty mixed-exponent classes of §2.6 — contains a row with $|\lambda_2| < 1$.*

Proof. Corollary 2.5 makes unequal exponents force a reducible cubic, and Theorem 5.1 forbids that when $|\lambda_2| < 1$. $\square$

Corollary 5.3 ([PROVED]). *Every row with $|\lambda_2| < 1$ has equal exponents at all three finite points, so its three finite fibres are Galois-conjugate and of one Kodaira type.*

This explains, rather than merely records, why the $|\lambda_2| < 1$ window is populated only by the equal-exponent classes.

Theorem 5.4 ([PROVED]). *If moreover $\lambda_2 \in \mathbb{R}$ (a real decayer) then χ is irreducible and totally real, $\Delta = 18adg - 4a^3g + a^2d^2 - 4d^3 - 27g^2 > 0$, and $|g| < |\lambda_1| \leq |a| + 2$.*

Proof. An irreducible cubic over $\mathbb{Q}$ has either three real roots or one real root and a complex-conjugate pair. In the second case the pair has equal modulus; if the real root were λ_2 then $|\lambda_1| = |\lambda_3|$, contradicting $|\lambda_2| \geq |\lambda_3|$ and $|\lambda_2| < |\lambda_1|$. So all three roots are real and $\Delta > 0$; finally $|g| = |\lambda_1||\lambda_2\lambda_3| < |\lambda_1|$. $\square$

Both statements were run against the 30 rows of Table 1 and against all 37088 integer cubics with $|\lambda_2| < 1$ in $|a| \leq 24$, $|d|, |g| \leq 70$: 0 reducible [VERIFIED: as stated]. Exactly one census row has $|\lambda_2| < 1$, namely $(154, 42, -128, -12, 8)$ with $|\lambda_2| = 0.45462\dots$, and its cubic is irreducible with $\Delta = -81\,920\,000 < 0$, so its two small roots are a complex pair and it is *not* a real decayer. The only other row with $|\lambda_2| \leq 1$, $(87, 24, -152, -16, 16)$, has $|\lambda_2| = 1$ exactly and a *reducible* cubic $(\lambda + 1)(\lambda^2 - 88\lambda - 64)$ — precisely the boundary case Theorem 5.1's proof leaves open. Consequently, in the four-term world a real decayer must live in an equal-exponent class with an irreducible totally real cubic.

The empirical side agrees. Over the whole extended window (all 27 equal-exponent classes, $|c| \leq 250$, $|f| \leq 500$, complete in (d, C) for every $a \leq 180$) there are exactly three rows with $|\lambda_2| < 1$ [VERIFIED: as stated]: the disguised I_0^* Padé row $(144, 32, 96, -32, 4)$ of class $(0; 2, -1, 3)$ ($|\lambda_2| = 0.35064$ real, $k = 1$, score $+0.0480$), the disguised $(36, 8, -48, 16, 4)$ of the same class $(0.65496, \text{non-real}, k = 1, -0.5768)$, and the genuine $(154, 42, -128, -12, 8)$ of $(0; 2, 1, 1)$ $(0.45462, \text{non-real}, k = 2, -1.2117)$. Over all 20 mixed-class candidates the smallest $|\lambda_2|$ is *exactly* 1, never below — Corollary 5.2 in action.

The third structural statement concerns what the disguise costs. The only integral three-term row in Zagier's sense whose Apéry limit involves Catalan's constant is Zagier **E**, $(n + 1)^2 u_{n+1} = (12n^2 + 12n + 4)u_n - 32n^2 u_{n-1}$ with $\lambda = (8, 4)$, $k = 2$, $\xi = \frac{1}{2}G$. By Proposition 3.1 its signed binomial transforms are integral four-term rows on the same local system — for $\nu = 1, 2, -1, -2$ they have $\lambda = \{8 - \nu, 4 - \nu, -\nu\}$, $k = 2$ and scores $-3.0986, -2.6931, -3.6094, -3.7918$, so the cusp move even improves the elementary score by up to $+0.69$ nats [VERIFIED: integrality to $n = 459$]. But:

Proposition 5.5 ([PROVED] and [VERIFIED: 70 digits]). *The signed binomial transform T of the companion is not the companion of the transformed row. Writing $L_0 B = t$ for Zagier **E** and $\tilde{y}(t) = (1 + \nu t)^{-1} y(t/(1 + \nu t))$ for the gauge, one has $L'(Ty)(t) = (L_0 y)(s)$ with $s = t/(1 + \nu t)$, so $L'(TB) = t/(1 + \nu t) \neq t$ and TB fails the four-term recurrence at every $n \geq 1$. The row's own companion W ($w_1 = 1$, $L'W = t$) is $W = TB + Ty$ with $(n + 1)^2 y_{n+1} = P_0 y_n - Q_0 y_{n-1} + \nu^n$, $y_0 = y_1 = 0$, and the exact Casoratian $(m + 1)^2 (a_m b_{m+1} - a_{m+1} b_m) = 32^m$ gives $\xi_\nu = \frac{1}{2}G \cdot A(\nu/32) - B(\nu/32)$ with $A = \sum a_n t^n$, $B = \sum b_n t^n$.*

The four values $\xi_1 = 0.48630684566125056749\dots$, $\xi_2 = 0.52037151700596857713\dots$, $\xi_{-1} = 0.43391612361971143068\dots$, $\xi_{-2} = 0.41311716537104535942\dots$ are all unidentified against the battery of §4.9 [VERIFIED: as stated], and structurally they must be, being values of the period and of its Eichler integral at an *interior* point of the disc; the control $\frac{1}{2}G$ itself is recovered by the battery at 79 digits. So the four-term world does contain integral rows built on the Catalan local system — the whole ν -family — but it is a four-point local system wearing a fifth point: no new arithmetic, and a canonical Apéry limit that is no longer $\frac{1}{2}G$. The contrast with §4.5, where the fold is a simple root of $\mathcal{R}$, the row is honestly four-term and the canonical limit is $\frac{1}{4}G$, is what makes the mixed-class rows a genuinely new realisation rather than a re-labelling.

6. THE SIX-POINT GLIMPSE: FIVE-TERM ROWS

One term further, the five-term row

$$(n + 1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1} + R(n)u_{n-2} - T(n)u_{n-3}, \quad u_0 = 1, u_{-1} = u_{-2} = u_{-3} = 0,$$

with $T = kn^2 + ln + m$ has an operator with *six* singular points and three accessory parameters.

Theorem 6.1 (five-term dictionary; [PROVED], [VERIFIED: as stated]). $L = \theta^2 - tP(\theta) + t^2Q(\theta + 1) - t^3R(\theta + 2) + t^4T(\theta + 3) = t^2\mathcal{R}D^2 + t\mathcal{S}D + t\mathcal{V}$ with

$$\mathcal{R} = 1 - at + dt^2 - gt^3 + kt^4,$$

$$\mathcal{S} = 1 - (a + b)t + (3d + e)t^2 - (5g + h)t^3 + (7k + l)t^4,$$

$$\mathcal{V} = -c + (d + e + f)t - (4g + 2h + j)t^2 + (9k + 3l + m)t^3.$$

The exponents are $(0, 0)$ at 0, $(0, \rho_i)$ at the four roots of $\mathcal{R}$ with $\rho_i = -T_c(t_i)/(t_i\mathcal{R}'(t_i))$, $T_c = \mathcal{S} - t\mathcal{R}' = 1 - bt + (d + e)t^2 - (2g + h)t^3 + (3k + l)t^4$, and $3 - s_1, 3 - s_2$ at ∞ with s_i the roots of T ; Fuchs reads $\sum_i \rho_i = s_1 + s_2 - 2$. All four exponents equal ρ iff $b = (1 - \rho)a$, $e = -2\rho d$, $h = -(1 + 3\rho)g$, $l = -(2 + 4\rho)k$,

and then $(j_1 + j_2)/M = s_1 + s_2 = 2 + 4\rho$ for $T(n) = C(Mn - j_1)(Mn - j_2)$. The Casoratian on four solutions is $W_{n+1} = -T(n)W_n/(n+1)^2$.

The free parameters are a, c, d, f, g, j, C : three accessory parameters $c = u_1$, $f = Q(0)$, $j = R(0)$ — exactly $6 - 3$ — plus four position moduli modulo the scaling. The scanner was validated against an exact brute force on a small box, both finding the same 17 integral rows, every one with a repeated characteristic root [VERIFIED: as stated].

6.1. The census and the disguise fraction. Seven classes were scanned over $a \leq 24$, $|c| \leq 12$, $|d| \leq 90$, $|f| \leq 40$, $|g| \leq 90$, $|j| \leq 60$, $|C| \leq 90$ in census mode, plus a deeper box on the productive class. In the all-semistable class $(0; 1, 1, 1)$, $T = C(n-1)^2$ (six I_n fibres), about 1700 integral rows are found, 99.6% with a repeated characteristic root, leaving **7** genuine six-point rows; in the gauge partner $(-1; 1, -1, -1)$ all 1608 integral rows have a repeated root, leaving **0** [VERIFIED: boxes as stated]. The pattern of §3.4 continues and intensifies: there the dominant degeneracy was the apparent singularity (98.2% disguised three-term rows); here it is a repeated characteristic root — the singular points collide and the object is not a six-point one at all — at 99.6%–100%. The gauge partner $(-1; 1, -1, -1)$ is the five-term analogue of $\mathbf{S}_{-1}$: empty, exactly as Proposition 2.7 makes $\mathbf{S}_{-1}$ empty. The scan of $(0; 1, 1, 1)$ was still running when it was stopped, so the list of seven is not claimed complete [OPEN].

(a, c, d, f, g, j, C)	λ	k	score	ξ
$(1, 0, -14, -2, 20, 8, -8)$	$2 \pm 2\sqrt{2}, -2, -1$	2	-2.6931	$\frac{\sqrt{2}}{3}L(g, 2)$
$(6, 2, 0, 0, -6, -2, -1)$	$3 \pm 2\sqrt{2}, -1, 1$	2	-2.0000	$\frac{\sqrt{2}}{3}L(g, 2)$ again
$(8, 2, -52, -8, 72, 24, -16)$	$12.5918, -2.3177, -2, -0.2741$	2	-2.8406	$0.2336058844421865619710 \dots$
$(4, 1, -13, -2, 9, 3, -1)$	$6.2959, -1.1588, -1, -0.1371$	2	-2.1474	$0.4672117688843731239421 \dots$
$(9, 3, 13, 2, 4, 1, -1)$	$7.2959, 1, 0.8629, -0.1588$	2	-2.0000	the same $0.46721176888 \dots$
$(5, 2, 0, 0, -20, -8, -16)$	$4, -2, 2, 1$	2	-2.6931	ξ^* (§4.8)
$(5, 2, 17, 3, 31, 12, 18)$	$1 \pm 2\sqrt{2}i, 2, 1$	2	-3.0986	none (complex pair)

$k = 2$ for all seven — the same free-integration phenomenon one term further — and *every period occurs at least twice*. Three of the seven have $|\lambda_2| = 1$ exactly, hence score -2.0000 .

6.2. A weight-3 cusp-form L -value as an honest Apéry limit. The first row of the list,

$$(n+1)^2 u_{n+1} = (n^2+n)u_n + (14n^2+2)u_{n-1} + (20n^2-20n+8)u_{n-2} + 8(n-1)^2 u_{n-3}, \quad (7)$$

$u_n = 1, 0, 4, 8, 40, 144, 616, 2544, 10936, 47200, \dots$, is integral [VERIFIED: exact rational arithmetic to $n = 400$; u_{400} has 271 decimal digits], primitive ($a = 1$), and has all $u_n \geq 0$ for $n \leq 400$. Its characteristic quartic is $\lambda^4 - \lambda^3 - 14\lambda^2 - 20\lambda - 8 = (\lambda+1)(\lambda+2)(\lambda^2 - 4\lambda - 4)$ with $\Delta = 2048 \neq 0$, so the six singular points $0, -1, -\frac{1}{2}, \frac{\sqrt{2}-1}{2}, -\frac{\sqrt{2}+1}{2}, \infty$ are distinct; the exponents are $(0, 0)$ at each of the five finite ones, so a logarithm is forced and none is apparent, and $(2, 2)$ at ∞ . Sharp $k = 2$ and score $= \log \frac{1}{2} - 2 = -2.6931$. Its Apéry limit is

$$\xi = 0.528749455595766833775958063859865871341 \dots = \frac{\sqrt{2}}{3}L(g, 2) = \frac{1}{2}\xi_{K3}(1) = \operatorname{Re} \xi_{K3}\left(\frac{5 \pm i\sqrt{2}}{27}\right)$$

[VERIFIED: 110 digits], where $L(g, 2)$ is the critical value of the weight-3 CM newform 32.3.d.a of Theorem 4.1.

This matters because the elliptic- $K3$ four-term row has a complex conjugate dominant pair and therefore no archimedean Apéry limit at all: there $L(g, 2)$ appears only as a fold constant, obtained by analytic continuation. In the six-point world the same L -value is an honest $\lim b_n/a_n$ of an integral row, with a score 0.955 nats better. The two local systems are not obviously the same — $\{5 \pm i\sqrt{2}, 1\}$ over $\mathbb{Q}(\sqrt{-2})$ against $\{2 \pm 2\sqrt{2}, -2, -1\}$ over $\mathbb{Q}(\sqrt{2})$, five points against six — but 32.3.d.a has CM by $\mathbb{Q}(\sqrt{-2})$ and level $32 = 2^5$, so both are 2-power-level objects and a correspondence is the obvious guess; identifying it is open [OPEN], as is a five-term analogue of the $\mathcal{J}$ -map test. Two further facts: $t = \infty$ has exponents $(2, 2)$ with a logarithm, so the operator has a *second MUM point*, whose characteristic quartic $(\lambda-1)(\lambda-2)(\lambda^2 - 2\lambda - 1)$ again has $|\lambda_2| = 2$ and whose slice contains no integral row [VERIFIED: $|c| \leq 40$,

$|f| \leq 60$, $|j| \leq 80$]; and $|\lambda_2| = 2$ exactly, so as a lattice engine the row's only candidate alignment prime is 2.

6.3. Periods repeat across worlds. Five independent numerical identities turned up, in two shapes. (A) A six-point Apéry limit equals the real part of a five-point fold constant: the row (7) against the elliptic $K3$; the row $(6, 2, 0, 0, -6, -2, -1)$ against the same $K3$ row, with score -2.0000 against its -3.6479 ; and the row $(8, 2, -52, -8, 72, 24, -16)$ against the mixed row $(\frac{1}{2}, 0; 3, 2, 4)_{r=27}$, $(23, 8, -216, -42, -324)$ — each to 110 digits — plus, inside the four-term world, the cross-row identity of §4.3 at 70 digits. (B) One Apéry limit shared by five rows in three classes and two worlds: the constant ξ^* of §4.8, to 319 digits.

Conjecture 6.2 ([CONJECTURE]). *The real part of a complex fold constant of a five-singular-point four-term row is, systematically, an honest archimedean Apéry limit $\lim b_n/a_n$ of an integral six-singular-point five-term row.*

The five instances are [VERIFIED: 70–319 digits]. The mechanism is presumably a quadratic base change: separating a complex-conjugate pair of singular points adds a puncture and replaces the fold constant by its $\text{Gal}(\mathbb{C}/\mathbb{R})$ -trace $\frac{1}{2}(\xi_+ + \xi_-)$. No period found in this work occurs only once, which suggests the census is seeing families rather than sporadic rows — but the identity is verified, not explained [OPEN].

7. ARITHMETIC: WHAT DOES NOT CHANGE

7.1. No new irrationality candidate. The *score* of a row is $\log(1/|\lambda_2|) - k$ with λ_2 the second largest characteristic root in modulus; a positive score is the archimedean part of an irrationality proof. Two complete sweeps settle the question in the four-term world.

Box W ($|\lambda_2| < 1$; $a \leq 300$ for 18 classes, $a \leq 400$ for $\mathbf{S}_0$) contains 79 rows with $|\lambda_2| < 1$, all in the classes $(\pm 1; 3, \dots)$ and $(\pm 1; 4, \dots)$, and *every one has a vanishing Frobenius obstruction*: a disguised three-term row.

Box PW ($|\lambda_2| < e^{-1}$, hence complete for a positive score whenever $k \geq 1$; all 27 classes, swept to completion for $a \leq 800$, $|c| \leq 60$, $|f| \leq 120$) contains *exactly two integral rows in total*, both in the class $(0; 2, -1, 3)$ with $\delta_\infty = 2$:

$$(144, 32, 96, -32, 4) : \lambda = 143.331, 0.3506, 0.3184, k = 1, \text{ score} = +0.0480,$$

$$(225, 50, -120, 40, 4) : \lambda = 225.532, -0.2662 \pm 0.0092i, k = 1, \text{ score} = +0.3229,$$

and *both have an exactly vanishing Frobenius obstruction at $t = \infty$* (the value is 0, not merely small); the exponents there are $(\frac{1}{2}, \frac{5}{2})$, so the monodromy is $-I$, an I_0^* fibre invisible to the projective local system, which therefore has only four singular points. They are the four-term avatars of the classical Legendre–Padé family, and $k = 1$ identifies them as Padé constructions.

Corollary 7.1 ([VERIFIED: boxes W and PW as stated]). *Over the whole scanned region — every Kodaira-admissible equal-exponent class, both windows — no genuine five-singular-point four-term row has a positive score. The only positive scores that occur belong to disguised three-term rows with $k = 1$, exactly as in the rigid case, where every positive score outside Apéry's and Beukers' rows has $k = 1$ and is a classical Legendre–Padé row. The four-term world produces no new irrationality candidate.*

Theorem 5.1 adds the structural half: a decaying row needs an irreducible characteristic cubic, so it cannot live in an unequal-exponent class, and by Theorem 5.4 a real decayer needs a totally real one. Neither condition is met anywhere in the scanned region.

7.2. The 2-adic content of the new Catalan rows. The six rows of §4.5 are *engines*, not decayers, so what matters for the two-row lattice machinery is their p -adic arithmetic. Write $\kappa_p = -\lim v_p(\text{den } b_n)/n$, $w_p = \lim v_p(\text{Cas}_n)/n$ with $\text{Cas}_n = a_n b_{n+1} - a_{n+1} b_n$, and $\sigma_p = w_p + 2\kappa_p$ the slope of $v_p(b_n/a_n - b_{n-1}/a_{n-1})$. Everything below is exact (rational or p -adic), archimedean at 1500 digits, measured to $n = 800$ for $p = 2$ and $n = 400$ for all p , with p swept to 71 [VERIFIED: as stated].

row	ξ_∞	g	κ_p	w_2	σ_2	slope primes $p \leq 71$	ξ_2
1	$\frac{1}{4}G$	-128	$\kappa_2 = -\frac{1}{2}$, else 0	$\frac{9}{2}$	$\frac{7}{2}$	$\{2\}$	$\frac{1}{4}\zeta_2(2)$
2	$\frac{1}{3}G - \frac{3}{16}\zeta(2)$	8	0	2	2	$\{2\}$	$\frac{1}{2}\zeta_2(2)$
3	$\frac{1}{8}\zeta(2) - \frac{1}{2}G$	32	0	$\frac{7}{2}$	$\frac{7}{2}$	$\{2\}$	$-\frac{1}{2}\zeta_2(2)$
4	$\frac{1}{32}(2\zeta(2) + 15L_{-3})$	32	0	3	3	$\{2\}$	$\frac{3}{8}L_2(2, \chi_{12})$
5	$\frac{1}{16}(15L_{-3} - 6\zeta(2))$	16	0	3	3	$\{2\}$	$\frac{3}{8}L_2(2, \chi_{12})$
6	$\frac{1}{16}(2\zeta(2) - 3L_{-3})$	-1	0	0	0	$\emptyset$	none
Zagier E	$\frac{1}{2}G$	32	0	5	5	$\{2\}$	$\frac{1}{2}\zeta_2(2)$

The 2-adic limits are exact to 1172–1200 2-adic digits, and the transfer $G \mapsto \zeta_2(2)$, $\zeta(2) \mapsto 0$, $L(2, \chi_{-3}) \mapsto \frac{4}{5}L_2(2, \chi_{12})$ is linear on the nose: rows 4 and 5 give $\frac{15}{32} \cdot \frac{4}{5} = \frac{3}{8}$ and $\frac{15}{16} \cdot \frac{4}{5} = \frac{3}{4}$, exactly as observed.

Three things are genuinely new in *form* [VERIFIED: as stated]. *Half-integral slopes*: rows 1 and 3 have $\sigma_2 = \frac{7}{2}$ and row 1 has $w_2 = \frac{9}{2}$, whereas every row in the earlier census has an integer $\sigma_p = v_p(c)$; a half-integral Casoratian slope means the two 2-adic Newton slopes of the pair (a_n, b_n) are conjugate over a *ramified* quadratic extension — here $\mathbb{Q}_2(\sqrt{2})$, so the irrationality of the singular points is visible 2-adically. *Decoupling*: $\sigma_2 = \frac{7}{2}, 2, \frac{7}{2}, 3, 3, 0$ against $v_2(g) = 7, 3, 5, 5, 4, 0$, so the three-term law $\sigma_p = v_p(c)$ fails for every one of the six and the slope must be measured; the 3×3 Casoratian still obeys $W_{n+1} = R(n)W_n/(n+1)^2$ exactly, but the 2×2 one has no closed form. *A negative κ_2* : row 1 is an integral row whose numerators are divisible by a growing power of 2, $v_2(a_n) = \frac{n}{2} + O(1)$, so $\kappa_2 = -\frac{1}{2}$ and $\sigma_2 = w_2 + 2\kappa_2 = \frac{9}{2} - 1 = \frac{7}{2}$ on the nose — the invariant $\sigma = w + 2\kappa$ confirmed in a regime it was never tested in.

But the *content* is old. The slope set is $\{2\}$ for rows 1–5 and empty for row 6 (checked to $p = 71$), exactly as for Zagier **E**, and the 2-adic values are the ledger’s own two constants with new rational scalars: row 2’s $\xi_2 = \frac{1}{2}\zeta_2(2)$ is literally **E**’s value with $\sigma_2 = 2$ instead of 5, and rows 4 and 5 sit in Zagier **F**’s 2-adic class but lack **F**’s second slope prime. The engine yield $\sum_p \sigma_p \log p - \log |\lambda_2|$ is 0.3466, 0.6931, 1.7329, 0.0693, 2.0101, 0 for rows 1–6, against 2.0794 for Zagier **E** and $\log 9 = 2.1972$ for **F**: *every one of the six is dominated by the engine it would replace*.

7.3. The Catalan ledger. The obstruction analysis [6] lists, as the evasion (E1) of the geometric wall, “a host with an irrational second singularity”: three-term rows cannot have one, their characteristic roots being rational hence integral, so $|t_2| \leq \frac{1}{4}$ at every placement and the conformal size is too small for the Hodge depth; rows with more singular points can. The six rows of §4.5 realise (E1) explicitly — irrational conjugate singularities, five punctures, periods in $\mathbb{Q}G + \mathbb{Q}\zeta(2)$ at 139 digits, full classical period matrices, best score +0.693 nats over Zagier **E** and equal to the best Catalan geometry previously known — and the ledger does not move: the 2-adic slope set is still $\{2\}$ and the limits still lie in the $\zeta_2(2)$ and $L_2(2, \chi_{12})$ classes, so the second wall (2.3) of [6], that the p -adic places cannot compensate because $G \in \mathbb{Q}$ is false at $p = 2$, is untouched. The Catalan class’s deficit is a missing $p = 3$ partner, and rows 1–3 supply $\{2\}$ again with a *smaller* 2-adic resource than **E**; meanwhile Theorem 5.1 forbids a real decayer in any unequal-exponent class. In the language of [6]: (E1) has been realised, and (2.3) has been re-decorated, not evaded. Whether the conditional-function machinery applies to rows whose fold is an order-2 orbifold point rather than a cusp is not settled here [OPEN].

Remark 7.2. The modular Catalan hosts themselves are not four-term rows: on the level-16 and $\Gamma_0(12)$ hosts the fold is a *double* root of the reversed characteristic polynomial, and the minimal recurrences have 6, 7 and 6 terms; the cusp move destroys their Apéry limits (Proposition D4 of `FOUR_TERM_DEEP.md`, $\xi_\nu = \frac{1}{2}G A(\nu/32) - B(\nu/32)$). So the four-term world and the modular Catalan world intersect only through the mixed-exponent rows of §4.

8. OPEN PROBLEMS

1. *Integrality of the sporadic rows: Zagier’s conjecture one step out.* For three-term rows integrality is explained by modular parametrisation, and the unconditional classification is exactly the statement that every integral row has one [4, 16]. Here 29 of the 30 rows of Table 1 fail the $\mathcal{J}$ -map test with $\deg \leq 18$, and no mechanism is known for their integrality; even for the one modular row, the elliptic

- $K3$ (6), integrality is experimental, and we do not know whether the index-16 group attaining the Riemann–Hurwitz bound is congruence [OPEN].
2. *Identify ξ^** (§4.8): one transcendental shared by five rows in three classes and two worlds, to 319 digits, unidentified against 145 classical constants. The natural next step is the fold constants of the $+\frac{1}{2}$ gauge and of the $(0, \mp\frac{1}{2}; 2, \cdot)$ classes, not computed here; if the pattern of §4.6 holds they should lie in $\mathbb{Q}\Gamma(\frac{1}{4})^4/\pi + \mathbb{Q} \cdot$ (something), which would pin ξ^* [OPEN].
 3. *Classify five-point elliptic surfaces with integral rows*. The configurations with five singular fibres form one-parameter families, so the locus they cut out in the $(\mathcal{I}_1, \mathcal{I}_2)$ -plane is a curve for each fibre type; computing those curves would give a genuine analogue of the cross-ratio finiteness of the rigid case, and would *predict* the $K3$ row instead of discovering it. The enumeration of the fifteen signatures of §2.7 into conjugacy classes of genus-zero subgroups of $\mathrm{PSL}_2(\mathbb{Z})$ — the analogue of the 28 groups in the four-point case — is also not carried out [OPEN].
 4. *Conjecture 6.2*: prove the base-change mechanism behind the five identities, and in particular explain why a six-point row over $\mathbb{Q}(\sqrt{2})$ has the period of a five-point elliptic $K3$ over $\mathbb{Q}(\sqrt{-2})$, and why the second such row does it with a score 1.65 nats better than the $K3$ row’s own [OPEN].
 5. *Are the six mixed rows a family?* The class $(-\frac{1}{2}, 0; 1, 0, 0)$ is complete over the box of §4.8 and contains exactly nine primitive rows, at $r = \pm 1, \pm 2, \pm 3, 4, 8, 8$, so within that box they are sporadic; whether they are the small members of a one-parameter family at larger $|r|$ or $|a|$ is open, and would change the picture, the three Catalan periods being three points of the two-dimensional space $\mathbb{Q}G + \mathbb{Q}\zeta(2)$ that a family would sweep out [OPEN].
 6. *Loose ends*: a closed form for $v_2(\mathrm{Cas}_n)$ of the six rows (measured as $w_2n + O(1)$ with the $O(1)$ in $[-3, 3]$, exact at multiples of 100) and a proof of $v_2(a_n^{(1)}) = \frac{n}{2} + v_2(a_n^{\mathbf{E}})$, observed at eleven checkpoints; a $\mathcal{J}$ -map test adapted to the mixed and five-term classes, none of which has been tested for modularity; the 21 equal-exponent Apéry limits and the 24 new fold periods, which are probably weight-two *multiple* polylogarithms at arguments in $\mathbb{Q}(\lambda_i)$, of which only the classical and depth-one parts were tested; the unequal-exponent classes with $\rho_1 \neq \rho_2 \neq \rho_3$, which need their own parametrisation; and completeness — box PW is complete for the positive-score question and box W for the decayer question in the stated classes and ranges, but no deep box is, and the cost law of §3.3 makes a literal $100\times$ extension of box G about 4×10^6 core-seconds [OPEN].

Reproducibility. All statements above are reproduced from the project documents in `consolidation/`: `FOUR_TERM_SCAN.md` (the dictionary, the classes, the census, the 30 rows, the $K3$ row’s discovery, Corollary 7.1); `FOUR_TERM_DEEP.md` (Theorems 2.9, 5.1, 5.4 and 6.1, the mixed classes, the six Catalan rows and their period matrices, the gauge partners, ξ^* , the 2-adic census, the five-term probe, Conjecture 6.2); `K3_ROW_PERIOD.md` for Theorem 4.1; and `CATALAN_OBSTRUCTION.md` for the ledger discussion of §7. Scripts, job files and logs — the scanners and their brute-force validators, the exact re-verification, the $\mathcal{J}$ -map test, the fold-constant machinery, the period batteries and the $\kappa_p/w_p/\sigma_p$ census — are under the directories `lattice/four_term/`, `lattice/four_term_deep/` and `lattice/k3_period/`; the output files carry every hit with its verdict and invariants.

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